# Hybrid Delay-Phase Precoding in Wideband UM-MIMO Systems Under True Time Delay and Phase Shifter Hardware Limitations

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***Abstract*— The exploitation of the substantial bandwidths available in the terahertz (THz) band has recently attracted considerable interest. However, beam squint effect is a significant obstacle in the design of wideband hybrid beamformers. The beam squint effect causes the radiation beam to deviate from the desired direction, resulting in substantial gain losses and hindering the effective utilization of available bandwidths. Delay-phase precoding (DPP), a combination of true time delay (TTD) elements and phase shifters (PSs) in the analog domain, has emerged as a potential solution to overcome the beam squint effect and maintain practical system design. However, existing hybrid precoding schemes that assume infinite resolution and unbounded range TTDs, as well as infinite resolution PSs, impose a significant burden in terms of hardware complexity and power consumption. In this paper, we introduce the Delta-Delay-Phase Precoding (DDPP) architecture, which significantly reduces the required delay range of TTDs, outperforming existing state-of-the-art solutions in the literature. Importantly, the proposed architecture maintains consistently superior performance, even as the number of antenna elements increases, making it scalable for UM-MIMO systems. Additionally, we propose hardware-aware designs for the hybrid analog precoder to combat beam squint effect while complying with the hardware limitations of TTDs and PSs. We formulate the design of TTDs and PSs as a joint optimization problem subject to their finite-resolution constraints. To tackle this non-convex optimization problem, we propose an iterative precoding algorithm based on alternating minimization. Simulation results demonstrate the superiority of the proposed hybrid analog precoding schemes over recent literature works. Particularly, the proposed precoding schemes achieve near-optimal performance despite the hardware limitations of TTDs and PSs.**

***Index Terms*— THz, UM-MIMO, beam squint, TTD, hybrid precoding, finite-resolution, limited-range, alternating minimization.**

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## I. Introduction

TERAHERTZ (THz) band, ranging between 0.1-10 THz, possesses enormous vacant bandwidths that could fulfill the demands of future bandwidth-hungry applications, such as virtual reality and holography [2], [3], [4], [5], [6], [7]. Several challenges must be addressed before reaping the benefits of this spectrum range. The spreading loss, which is proportional to the square of the carrier frequency, is a significant obstacle that hinders the utilization of THz frequencies. Additionally, molecular absorption causes attenuation that further increases the path loss in the THz band [8]. As a result, numerous efforts have been dedicated to overcome the significant losses associated with THz frequencies and extending the communication range [9], [10], [11], [12].

As a result, ultra-massive multiple input multiple output (UM-MIMO) technology has become a primary focus of research activities as a key-enabler for THz communications. Due to the submillimeter-level wavelengths of signals in the THz band, the size of antenna elements is proportional to the wavelength, allowing for several thousands of antenna elements to be packed into a relatively small area [13]. This results in high gains as a result of constructive interference between signals radiated from individual antenna elements, known as beamforming gain [14].

Beamforming can be achieved through different methods, including digital, analog, and hybrid beamforming. However, digital beamforming with a dedicated RF chain per antenna element is not practical due to the high power consumption of the huge number of RF chains and their associated digital to analog converters (DACs) [15]. Analog beamforming, on the other hand, utilizes a single RF chain but lacks the potential to multiplex different data streams, which has led to the emergence of hybrid beamforming as a compromise solution with a few number of RF chains [16], [17]. Conventional hybrid beamformers use a phase shifter (PS) network to adjust the phases of signals before radiation, with the weights of PSs typically designed for the carrier frequency and applied to the entire bandwidth. In narrowband communication systems, this approach has a negligible effect due to the convergence of frequencies. However, in wideband communication systems, the beam squint effect causes the radiation angle to vary at each frequency within the bandwidth, leading to a frequency-dependent radiation angle [18]. This becomes more severe with UM-MIMO systems, where the radiated beams are very sharp and even a slight squint can have a significant impact [19]. Therefore, frequency-dependent phase shifts are required to achieve a frequency-independent radiation angle, which is what a true time delay (TTD) element can provide [20].

TTD elements have been proposed by several studies in the literature to compensate for the beam squint effect. One direct implementation of TTD elements is to replace the PS network with a TTD network, such that each antenna element is preceded by a dedicated TTD element [21]. However, such implementation may not be energy-efficient, considering that the power consumption of a TTD element is almost three times that of a conventional PS [22], [23]. To this end, a hybrid analog precoder consisting of a TTD network and a PS network has been proposed, where the PS network is driven by a limited number of TTDs [24], [25], [26]. These works often rely on impractical assumptions, such as unlimited delay range and infinite-resolution TTDs, as well as infinite-resolution PSs. However, practical TTD fabrications at THz band can achieve only limited-range and finite-resolution time delays. These practical constraints will be discussed in the subsequent subsection.

### A. Recent Advances and Limitations of TTD Technologies in the THz Band

The authors in [27] provide a comprehensive overview of recent TTD circuit realizations operating above 100 GHz. The first such TTD electronic circuit is introduced in [28], designed in 0.13 $\mu$m silicon-germanium (SiGe) BiCMOS technology. It employs switchable transmission lines to achieve a maximum delay range of 6.64 ps with a time delay resolution of 0.44 ps. This design is adjustable so one can reconfigure it to obtain different maximum delay line and resolution. However, this flexibility comes at the cost of a trade-off between delay line and resolution. This TTD fabrication is adopted in the design of a phased-array transceiver, that demonstrates a data rate up to 200 Gbps at THz band [29]. Similarly, the work in [30] presents a TTD design in 0.13 $\mu$m SiGe BiCMOS technology, operating above 200 GHz and achieving a limited delay range of 1.47 ps with a time delay resolution of 0.39 ps. Moreover, the work in [31] introduces a TTD circuit based on cascaded delay units, designed in 35-nm Indium Gallium Arsenide Metamorphic High-Electron-Mobility Transistor (InGaAs mHEMT) technology, operating between $220 - 325$ GHz. This design achieves a bounded maximum delay range of 3.398 ps with a time delay resolution of 0.227 ps. These studies collectively indicate that research in TTD realizations for the THz band is still nascent, with only limited delay ranges achievable. It is evident from [28], [29], [30], and [31] that TTD design involves trade-offs among various factors such as delay range, resolution, insertion loss and footprint area [27]. For example, the TTD chip introduced in [28] can be reconfigured to achieve a larger maximum delay range of 14.272 ps but at the cost of a coarser time resolution of 3.568 ps. Given these limitations, it is imperative to consider the current state-of-the-art TTD capabilities in the THz band when designing hybrid precoders.

### B. Problem Statement and Related Work

Hardware constraints, such as the maximum delay range and time delay resolution of TTDs, along with the resolution of PSs, limit the hybrid analog precoder's performance. Achieving long-range TTDs requires lengthy transmission lines, resulting in higher insertion loss and increased silicon footprint area [27]. High-resolution TTDs demand cascading fine-time delay units, leading to more complex hardware, greater power consumption and larger silicon area requirements [30], [32]. PSs also have finite resolution and are digitally controlled, necessitating their inclusion in considerations for hybrid precoding schemes [32].

Recent literature, such as [24] and [25], proposed hybrid precoding schemes that neglect these hardware limitations, assuming impractically unlimited-range and infinite-resolution TTDs and PSs. Some efforts, like the work in [33], consider the maximum delay range constraint of TTDs but still assume infinite resolution delay elements. Although this approach outperforms the DPP scheme [25] in terms of array gain, it lacks scalability; the array gain degrades as the number of antenna elements increases. Furthermore, [33] also assumes infinite-resolution TTDs and PSs, a constraint that is impractical as previously discussed. The work in [34] aims to improve energy efficiency through a dynamic sub-array with fixed true-time-delay architecture but still requires long-range TTD elements. Therefore, it is crucial to account for practical, finite-resolution and limited-range TTD elements, as well as finite-resolution PSs, in the design of hybrid analog precoders.

### C. Contributions

In this work, we propose practical hybrid analog precoding schemes, which combat beam squint effect at one hand and comply with the hardware limitations of TTDs and PSs on the other hand. The main contributions of this work are summarized as follows.

- We propose the delta-delay-phase precoding (DDPP) architecture to enable wideband beamforming while considering the practical limitations of limited-range and time resolution of TTDs. In contrast to the conventional delay-phase precoding architecture [25], DDPP inserts TTD elements between successive subarrays so that the time delay applied to a signal in a particular subarray is achieved by a group of TTD elements instead of a dedicated TTD element. Additionally, the proposed scheme addresses time resolution constraints.
- We propose a hybrid practical precoding scheme, which jointly optimizes the TTD network and PS network considering their hardware limitations. Specifically, we formulate the design of the hybrid analog precoder as a non-convex joint optimization problem subjected to the resolution constraints of TTDs and PSs and also to the limited-range of TTDs. An alternating minimization is utilized to solve the optimization problem.
- To reduce the computational complexity of the hybrid practical precoding scheme, we propose a low-complexity precoding approach, which ensures a fixed value for all

TTDs of the system, thereby significantly reducing the computational overhead compared to the original hybrid practical precoding scheme.
- We assess the power consumption of the proposed DDPP architecture and analyze the computational complexity of the proposed hybrid precoding schemes with respect to both the fractional bandwidth and the number of antenna elements.
- We carry out extensive simulations to verify the effectiveness of the proposed precoding architecture and the hybrid precoding schemes in dealing with the hardware limitations of TTDs and PSs compared to the state-of-art hybrid precoding approaches in the literature.

In this context, a preliminary investigation of the hybrid analog precoding under the hardware limitations of TTDs is presented in [1]. Our work in [1] primarily addresses the finite-resolution limitation of TTDs in the design of the hybrid analog precoder, whereas the maximum delay range of TTDs is considered as a constraint to the optimization problem. Nevertheless, the work in [1] suffers array gain loss as the maximum delay range supported by TTDs decreases. In the same vein, a remarkable gain loss occurs with the adoption of higher number of antenna elements, which makes the work in [1] unscalable for UM-MIMO systems. Moreover, the work in [1] does not consider the hardware limitations of PSs, where impractical infinite-resolution PSs are assumed.

The remainder of this paper is organized as follows. In Sec. II, we present the wideband channel model and an overview of beam squint effect and recent hybrid precoding approaches. In Sec. III, we introduce the proposed delta-delay-phase precoding (DDPP) architecture and present the time resolution constrained hybrid precoding (TR-HP). In Sec. IV, we propose the hybrid practical precoding (HPP), which accounts for the hardware limitations of TTDs and PSs. In Sec. V, we propose the fixed-delta hybrid practical precoding (FD-HPP), that features low computational complexity. In Sec. VI, we present the power and computational metrics of the proposed schemes. Simulation results and discussions are presented in Sec. VII before concluding the paper in Sec. VIII.

*Notation*: $\mathbf{A}$, $\mathbf{a}$, and $a$ denote a matrix, a vector, and a scalar, respectively. $\mathbf{a}(i)$ is the $i^{\text{th}}$ element of a vector $\mathbf{a}$ and $\mathbf{A}(i,j)$ is the element with the indicies $i$ and $j$ of a matrix $\mathbf{A}$ whereas $\mathbf{A}(i:j,:)$ is a submatrix of $\mathbf{A}$ from its $i^{\text{th}}$ row to the $j^{\text{th}}$ row. $[\mathbf{a} \mid \mathbf{b}]$ indicates the concatenation of $\mathbf{a}$ and $\mathbf{b}$. $\|\mathbf{A}\|_F$, $\mathbf{A}^*$,$\mathbf{A}^T$, $\mathbf{A}^H$ denote the Frobenius norm, conjugate, transpose, and Hermitian of a matrix $\mathbf{A}$, respectively. $\mathbf{blkdiag}(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)$ represents a block diagonal matrix, whose blocks are $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$. $\mathbf{0}_{n,m}$ and $\mathbf{1}_n$ denote an $n \times m$ empty matrix and an $n$ long all-ones vector, respectively. $\mathbf{I}_n$ denotes the $n \times n$ identity matrix. $\mathbf{QZ}_{\Upsilon}(x)$ function quantizes a real number $x$ to the nearest element in set $\Upsilon$. $\mathrm{Re}\{z\}$ and $\mathrm{ang}\{z\}$ return the real part and the angle of a complex number $z$, respectively. $\lfloor a \rceil$ rounds a real number $a$ to the nearest integer and $\mathrm{sign}(a)$ returns the sign of $a$. $\min(a,b)$ returns the minimum of real numbers $a$ and $b$. $\mathrm{Ln}(z)$ is the principle value for the natural logarithm $\ln(z)$ of a complex number $z$. $\preceq$ indicates the element-wise vector inequality. $\mathbb{C}$, $\mathbb{R}$, $\mathbb{Z}$, and $\mathbb{B}$ are the sets of complex numbers, real numbers, integers, and boolean, respectively.

## II. Wideband Channel Model and Beam Squint Effect

### A. Wideband Channel Model

We consider the downlink of a wideband UM-MIMO-OFDM system with $M$ subcarriers such that the frequency of the $m^{\text{th}}$ subcarrier is given by $f_m = f_c + \frac{B}{M}[m - \frac{M+1}{2}]$, where $f_c$ and $B$ indicate the carrier frequency and bandwidth, respectively. We denote $\mathbf{f} = [f_1, \ldots, f_M]^T \in \mathbb{R}^{M \times 1}$ as the subcarriers' vector. The base station (BS) is equipped with a uniform linear array (ULA) with $N_{\mathrm{T}}$ antenna elements serving $U$ single-antenna mobile users. The communication channel is modeled using a geometrical approach, which accounts for the presence of $L$ scatterers between the BS and the mobile users. Hence, the channel can be modeled as a sum of $L$ distinct propagation paths such that the channel vector between the BS and the $u^{\text{th}}$ mobile user at the $m^{\text{th}}$ subcarrier is expressed as

$$\mathbf{h}_{m,u} = \sqrt{\frac{N_T}{L}} \sum_{l=1}^{L} \alpha_{l,u} e^{-j2\pi\gamma_{l,u} f_m} \mathbf{a}^H(\Theta_m^{l,u}). \tag{1}$$

$\alpha_{l,u}$ and $\gamma_{l,u}$ are the complex path gain and path delay of the $l^{\text{th}}$ path for the $u^{\text{th}}$ user, respectively. $\mathbf{a}(\Theta_m^{l,u})$ is the response vector of the phased array at the $m^{\text{th}}$ subcarrier, where $\Theta_m^{l,u}$ denotes the spatial frequency as follows

$$\Theta_m^{l,u} = \frac{2\pi f_m d}{v} \sin\theta_{l,u}, \tag{2}$$

where $d$, $v$, and $\theta_{l,u} \in [-\pi/2, \pi/2]$ are the array interelement distance, the speed of light and the angle of departure (AoD) of the $l^{\text{th}}$ path for the $u^{\text{th}}$ user, respectively. The response vector of the phased array for a ULA at the $m^{\text{th}}$ subcarrier is given as

$$\mathbf{a}(\Theta_m^{l,u}) = \frac{1}{\sqrt{N_T}}[1, e^{j\Theta_m^{l,u}}, \ldots, e^{j(N_T-1)\Theta_m^{l,u}}]^T. \tag{3}$$

In conclusion, the channel matrix $\mathbf{H}_m \in \mathbb{C}^{U \times N_T}$ between the BS and the $U$ mobile users is expressed as

$$\mathbf{H}_m = [\mathbf{h}_{m,1}^T, \ldots, \mathbf{h}_{m,U}^T]^T. \tag{4}$$

### B. Beam Squint Effect

Beam squint effect occurs as a consequence of using frequency-independent PSs in the analog part of wideband hybrid analog beamformers, which is denoted as PS-based precoding and described as

$$\mathbf{a}(\Theta_c^{l,u}) = \frac{1}{\sqrt{N_T}}[1, e^{j\Theta_c^{l,u}}, \ldots, e^{j(N_T-1)\Theta_c^{l,u}}]^T, \tag{5}$$

where $\Theta_c^{l,u}$ indicates the spatial frequency at $f_c$. From (5), we note that the response vector of the phased array is designed for the carrier frequency $f_c$ and applied to the whole bandwidth. In consequence, margin subcarriers suffer array gain losses in comparison with the carrier frequency, where

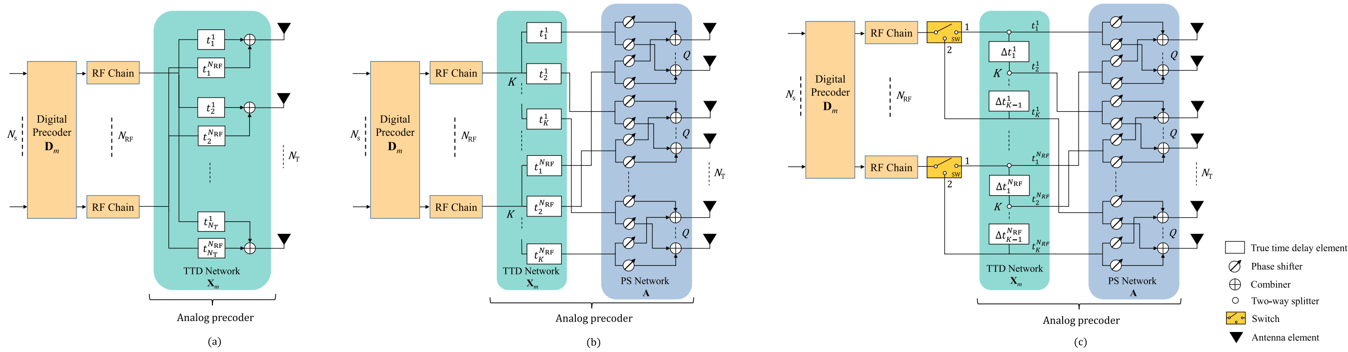


Fig. 1. Wideband hybrid precoding: (a) full-TTD precoding architecture (b) delay-phase precoding (DPP) architecture (c) the proposed delta-delay-phase precoding (DDPP) architecture.

the normalized array gain at the $m^{\text{th}}$ subcarrier for a given angle $\theta_{l,u}$ is expressed as

$$g_m(\theta_{l,u}) = |\mathbf{a}^H(\Theta_c^{l,u})\ \mathbf{a}(\Theta_m^{l,u})| = \frac{1}{N_{\mathrm{T}}}\left|\sum_{n=1}^{N_{\mathrm{T}}} e^{j(n-1)(\Theta_m^{l,u}-\Theta_c^{l,u})}\right|. \quad (6)$$

From (6), it is pretty clear that $g_m(\theta_{l,u}) = 1$ when $\Theta_m^{l,u} = \Theta_c^{l,u}$, namely when $f_m = f_c$. Meanwhile, $g_m(\theta_{l,u}) < 1$ when $f_m \neq f_c$. As a result, the PS-based beamformer will suffer gain losses, that hinders a full utilization of the huge available bandwidths at THz band. To address this, TTD elements have been introduced in [20] in order to provide frequency-dependent phase shifts and thus eliminate the beam squint effect. Next, we will investigate the conditions that dictate when TTD elements are essential for mitigating the beam squint effect.

### *C. When Are TTD Elements Essential for Beam Squint Mitigation?*

In this subsection, we establish the boundary conditions that determine when TTD elements are necessary to mitigate the beam squint effect and when they are not.

By substituting (2) in (6), the normalized array gain of a PS-based analog beamformer is given as

$$g_m(\theta_{l,u}) = \frac{1}{N_{\mathrm{T}}}\left|\sum_{n=0}^{N_{\mathrm{T}}-1} e^{jn2\pi\frac{d}{v}(f_m-f_c)\sin\theta_{l,u}}\right|. \quad (7)$$

Knowing that $\sum_{n=0}^{N-1} e^{jnx} = \frac{\sin(Nx/2)}{\sin(x/2)} e^{jx(N-1)/2}$, we can formulate

$$g_m(\theta_{l,u}) = \frac{1}{N_{\mathrm{T}}}\left|\frac{\sin[N_{\mathrm{T}}\pi\frac{d}{v}(f_m-f_c)\sin\theta_{l,u}]}{\sin[\pi\frac{d}{v}(f_m-f_c)\sin\theta_{l,u}]} .e^{j(N_{\mathrm{T}}-1)\pi\frac{d}{v}(f_m-f_c)\sin\theta_{l,u}}\right|. \quad (8)$$

Accordingly, the normalized array gain at the $m^{\text{th}}$ subcarrier is given by

$$g_m(\theta_{l,u}) = \frac{1}{N_{\mathrm{T}}}\left|\frac{\sin[N_{\mathrm{T}}\pi\frac{d}{v}(f_m-f_c)\sin\theta_{l,u}]}{\sin[\pi\frac{d}{v}(f_m-f_c)\sin\theta_{l,u}]}\right| = \left|\mathrm{Diric}_{N_{\mathrm{T}}}\left[2\pi\frac{d}{v}(f_m-f_c)\sin\theta_{l,u}\right]\right|, \quad (9)$$

where $\mathrm{Diric}_n(x) = \frac{\sin(nx/2)}{n\sin(x/2)}$ denotes the sinc Dirichlet function. We consider that a PS-based analog beamformer is sufficient provided that all subcarriers fall within the 3-dB bandwidth, i.e., $g_m(\theta_{l,u}) \geq 0.5\ \forall\ m = 1, 2, \ldots, M$. Consequently, a PS-based beamformer is sufficient when $g_M(\theta_{l,u}) \geq 0.5$, as the last subcarrier is most affected by the beam squint effect. The normalized array gain of the $M^{\text{th}}$ subcarrier is given as follows

$$g_M(\theta_{l,u}) = \left|\mathrm{Diric}_{N_{\mathrm{T}}}\left[2\pi\frac{d}{v}(f_M-f_c)\sin\theta_{l,u}\right]\right|. \quad (10)$$

By substituting $f_M - f_c = B/2$ and $d = \lambda_c/2$, (10) can be written as

$$g_M(\theta_{l,u}) = \left|\mathrm{Diric}_{N_{\mathrm{T}}}\left[\frac{\pi}{2}\beta\sin\theta_{l,u}\right]\right|, \quad (11)$$

where $\beta = \frac{B}{f_c}$ denotes the fractional bandwidth. To gain insight on the effect of the number of antenna elements and fractional bandwidth over different angles, the normalized array gain for the $M^{\text{th}}$ subcarrier per multiple users is given by

$$\bar{g}_M = \frac{1}{U}\sum_{u=1}^{U}\left|\mathrm{Diric}_{N_{\mathrm{T}}}\left[\frac{\pi}{2}\beta\sin\theta_{l,u}\right]\right|. \quad (12)$$

Fig. 2 depicts $\bar{g}_M$ versus the number of antennas $N_{\mathrm{T}}$ and the fractional bandwidth $\beta$. The boundary line, representing the condition where $\bar{g}_M = 0.5$, divides the figure into two regions: a large area where $\bar{g}_M < 0.5$ (above the boundary) and a smaller one where $\bar{g}_M > 0.5$ (below the boundary). Notably, as either the number of antennas or the fractional bandwidth increases, the normalized array gain for the $M^{\text{th}}$ subcarrier drops below 0.5. This decrease suggests that the PS-based beamformer becomes inadequate in countering the beam squint effect. For instance, with just a fractional bandwidth as low as $\beta = 0.05$, the gain falls below 0.5 when $N_{\mathrm{T}} > 96$. Conversely, even with a limited number of antennas, i.e. $N_{\mathrm{T}} = 32$, the gain drops below 0.5 when $\beta > 0.1$. This evidence strongly advocates for the necessity of a TTD-aided beamformer to address the beam squint effect, particularly in scenarios located in the expansive region above the boundary. In THz communications, which often involve large-scale arrays with many antenna elements [13], [35], we anticipate that realistic operating scenarios will predominantly

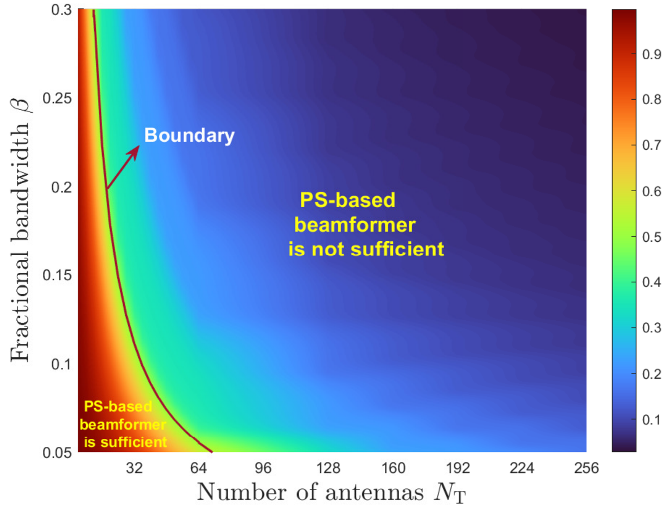


Fig. 2. Normalized array gain at the $M^{\text{th}}$ subcarrier per multiple users versus the number of antennas $N_{\mathrm{T}}$ and the fractional bandwidth $\beta$.

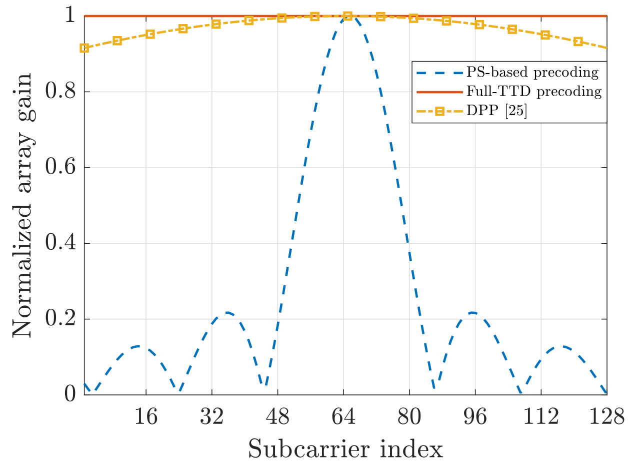


Fig. 3. Normalized array gain versus subcarriers when $N_{\mathrm{T}} = 256$.

be situated in the expansive region, even at low fractional bandwidths. In this region, the use of TTD-aided beamforming becomes essential for countering the beam squint effect.

### D. TTD-Aided Hybrid Analog Beamforming

TTD elements provide frequency-dependent phase shifts, which can mitigate the beam squint effect. The optimal analog precoder required to combat squint effect over all subcarriers is given as

$$\mathbf{C}_m^l = [\mathbf{c}_m^{l,1}, \ldots, \mathbf{c}_m^{l,U}], \tag{13}$$

where $\mathbf{c}_m^{l,u} = \mathbf{a}(\Theta_m^{l,u})$ represents the optimal analog precoder over the $l^{\text{th}}$ path for the $u^{\text{th}}$ user at the $m^{\text{th}}$ subcarrier. As a result, the optimal analog precoder is capable of achieving a unity array gain of $g_m(\theta_{l,u}) = |\mathbf{a}^H(\Theta_m^{l,u})\ \mathbf{a}(\Theta_m^{l,u})| = 1$ across all subcarriers, thereby eliminating the beam squint effect. However, implementing this precoder necessitates delay-controlled analog precoding instead of the conventional phase-controlled analog precoding. This entails embedding a dedicated TTD per antenna element in the RF front-end to enable wideband beamforming, as illustrated in Fig. 1.a, which is denoted in this paper as full-TTD precoding. The implementation of full-TTD precoding can be impractical in terms of power consumption and hardware complexity, especially when a large number of TTDs are required. In consequence, DPP has been introduced as a compromise between performance and hardware complexity [25].

DPP technique employs a limited number of TTDs, where the antenna array is divided into $K$ subarrays, each with $Q = N_{\mathrm{T}}/K$ antenna elements controlled by a single TTD element as depicted in Fig. 1.b. This configuration provides a practical system design while still approaching near-optimal performance, comparable to that of a full-TTD precoder.

Fig. 3 displays the normalized array gain computed using (6) for the various precoding schemes mentioned earlier. The computation assumes half-wavelength spaced antennas with $f_c = 300$ GHz, $B = 30$ GHz, $M = 128$, $K = 16$, $N_{\mathrm{T}} = 256$, and $\theta_{l,u} = 0.5$ rad. The results show that the full-TTD precoder achieves a constant normalized gain across all subcarriers, while the normalized gain of the PS-based precoder decreases as the subcarrier frequency moves away from the carrier frequency. In contrast, DPP demonstrates near-optimal performance with significantly reduced hardware complexity compared to the full-TTD precoder.

DPP requires time delay values that scale up with the TTD's index. Specifically, the time delay value of the $k^{\text{th}}$ TTD belonging to the $n^{\text{th}}$ RF chain is given as [25]

$$t_k^n = -(k-1)Q\frac{d}{v}\sin\theta_{l,u} \quad \forall\ k = 1, 2, .., K. \tag{14}$$

For instance, the required time delay range is nearly 400 ps when $N_{\mathrm{T}} = 256$, $K = 16$, and $\theta_{l,u} = \pi/2$. This time delay is considerably higher than the time delay range attainable with state-of-the-art TTDs at THz band, i.e. 6.67 ps [28]. The authors in [33] proposed joint DPP that satisfies the maximum time delay constraint of TTDs. This approach outperforms DPP [25] under the adoption of limited-range TTDs. However, the performance of this approach deteriorates as the number of antennas increases, which make it unsuitable for UM-MIMO systems. Moreover, this work assumes impractical infinite-resolution TTDs and PSs. However, in practical wideband communication systems, practical low-resolution TTDs and PSs are more appropriate to balance energy efficiency and performance [36]. Building upon the architecture proposed in [37] for angular coverage expansion, we propose next the delta-delay-phase precoding architecture that guarantees scalability for UM-MIMO systems, even with practical finite-resolution and limited-range TTDs.

## III. Delta-Delay-Phase Precoding Under True Time Delay Constraints

In this section, we present the proposed precoding architecture, which enables wideband beamforming in UM-MIMO systems using practical TTDs with limited range and finite resolution.

### A. The Proposed System Model

The required phase shift between adjacent subarrays in order to combat beam squint effect at the $m^{\text{th}}$ subcarrier is given as [38]

$$\Delta\vartheta_m = Q\frac{2\pi d}{\lambda_m}\sin\theta_{l,u}, \tag{15}$$

where $\lambda_m$ is the wavelength of the $m^{\text{th}}$ subcarrier. Consequently, the required time delay difference between adjacent subarrays is derived by matching the phase shift obtained from a TTD element $\psi_m$ to that in (15) as follows

$$\psi_m = -2\pi f_m \Delta t_k^n = -\frac{2\pi v}{\lambda_m}\Delta t_k^n \triangleq Q\frac{2\pi d}{\lambda_m}\sin\theta_{l,u}, \tag{16}$$

where $\Delta t_k^n$ denotes the required time delay difference between the $(k+1)^{\text{th}}$ and $k^{\text{th}}$ subarrays of the $n^{\text{th}}$ RF chain. Consequently, the required time delay difference between adjacent subarrays of a given RF chain is given as

$$\Delta t_k^n = -Q\frac{d}{v}\sin\theta_{l,u} \quad \forall\ k = 1,2,.,K-1. \tag{17}$$

Observing (17), we note that the time delay difference between adjacent subarrays remains constant, provided that they have the same size, i.e., the same number of antenna elements $Q$. Building on this observation, we propose a novel architecture for hybrid analog precoding, where TTDs are inserted between neighboring subarrays, as illustrated in Fig. 1.c. We refer to this proposed architecture as the delta-delay-phase precoding (DDPP) architecture, where the time delay difference between adjacent subarrays serves as the key optimization variable of interest in the TTD network. The RF signal is fed to successive subarrays through two-way splitters with cumulative time delays. Particularly, the time delay value applied on the signal fed to the $k^{\text{th}}$ subarray of the $n^{\text{th}}$ RF chain is realized by a set of TTDs $t_k^n = \sum_{i=1}^{k-1}\Delta t_i^n = \Delta t_1^n + \Delta t_2^n + \ldots + \Delta t_{k-1}^n$ rather than by a dedicated TTD as in [25], where $\Delta t_i^n$ as a function of the time delay values over subarrays is given as

$$\Delta t_i^n = t_{i+1}^n - t_i^n \quad \forall\ i \in 1,2,.,K-1. \tag{18}$$

In contrast, the proposed DDPP architecture aims at alleviating the maximum time delay limitation of TTDs. Specifically, the DDPP architecture can surpass the maximum time delay of a single TTD element, by utilizing multiple TTDs in a stepwise manner. By implementing time delays between adjacent subarrays, the maximum delay limitation is imposed on the inter-subarray time delay difference instead of the time delay needed per subarray, thereby alleviating the burden on TTDs. Specifically, in comparison with (14), the proposed architecture achieves a reduction in the maximum delay range by a factor of $K-1$. Accordingly, we define $\mathbf{t}_n^{\triangle} = [\Delta t_1^n, \Delta t_2^n, \ldots, \Delta t_{K-1}^n] \in \mathbb{R}^{K-1\times 1}$ as the time delay vector of TTDs belonging to the $n^{\text{th}}$ RF chain of the DDPP architecture. Meanwhile, $\mathbf{t}_n = [t_1^n, t_2^n, \ldots, t_K^n] \in \mathbb{R}^{K\times 1}$ denotes the resulting time delays over subarrays such that $\mathbf{t}_n \preceq \mathbf{1}_K t_{\text{req}}$, where $t_{\text{req}} = A_{\text{phys}}/v$ and $A_{\text{phys}}$ indicate the maximum required time delay over subarrays and the array physical aperture, respectively [39]. Consequently, as the time delay difference between adjacent subarrays is fixed as depicted in (17), the time delay vector of TTDs belonging to the $n^{\text{th}}$ RF chain is expressed as

$$\mathbf{t}_n^{\triangle} = -Q\frac{d}{v}\sin\theta_{l,u}\mathbf{1}_{K-1}. \tag{19}$$

As demonstrated in (19), the values of $\mathbf{t}_n^{\triangle}$ depend on $\theta_{l,u}$ and can be either positive or negative. However, TTD elements can not generate negative time delays. Thus, we propose inserting a 2-output switch between each RF chain and its corresponding analog precoder. This switch routes the signal to the first subarray through its $1^{\text{st}}$ output when the elements of $\mathbf{t}_n^{\triangle}$ are positive, which occurs when $\theta_{l,u} \in [-\pi/2, 0]$. It rather closes the $2^{\text{nd}}$ output feeding the signal to the last subarray first in case the elements of $\mathbf{t}_n^{\triangle}$ are negative, which occurs when $\theta_{l,u} \in [0, \pi/2]$. We model this switch as a binary selecting vector $\mathbf{e}$ as follows

$$\mathbf{e}(\theta_{l,u}) = \frac{1}{2|\sin(\theta_{l,u})|}\begin{bmatrix}|\sin(\theta_{l,u})| + \sin(\theta_{l,u})\\ |\sin(\theta_{l,u})| - \sin(\theta_{l,u})\end{bmatrix}. \tag{20}$$

The delta-based time delay vector over subarrays of the $n^{\text{th}}$ RF chain when $\theta_{l,u} \in [-\pi/2, 0]$ is given as

$$\mathbf{t}_n^{+} = \left[0, \Delta t_1^n, \ldots, \sum_{k=1}^{K-1}\Delta t_k^n\right]^T, \tag{21}$$

where the first element of $\mathbf{t}_n^{+}$ is zero, since the RF-signal is fed to the first subarray with no time delay in the proposed DDPP architecture. Likewise, the delta-based time delay vector over subarrays of the $n^{\text{th}}$ RF chain when $\theta_{l,u} \in [0, \pi/2]$ is given as $\mathbf{J}\mathbf{t}_n^{+}$, where $\mathbf{J} \in \mathbb{B}^{K\times K}$ denotes the permutation matrix with antidiagonal elements set to one and all other elements set to zero. Consequently, the resulting time delays over subarrays of the $n^{\text{th}}$ RF chain is given as

$$\mathbf{t}_n = [\mathbf{t}_n^{+} \quad \mathbf{J}\mathbf{t}_n^{+}]\mathbf{e}(\theta_{l,u}). \tag{22}$$

The proposed DDPP architecture adopts hybrid precoding, where a PS network is embedded in the analog precoder as shown in Fig. 1.c. We assume that the PS network is designed for the carrier frequency using (5). Consequently, the total phase difference between adjacent subarrays resulting from the TTD network and the PS network at the $m^{\text{th}}$ subcarrier is expressed as

$$\tilde{\Delta}\vartheta_m = \underbrace{Q\frac{2\pi d}{\lambda_m}\sin\theta_{l,u}}_{\text{(a)}} + \underbrace{Q\frac{2\pi d}{\lambda_c}\sin\theta_{l,u}}_{\text{(b)}}. \tag{23}$$

We note from (23) that the phase difference between adjacent subarrays is composed of two terms: (a) and (b). Particularly, term (a) in (23) represents the frequency-dependent phase difference required to combat beam squint effect, whereas term (b) denotes a fixed phase difference delivered by the PS network. Aiming at realizing the required phase shift difference between adjacent subarrays given in (15), we need to compensate term (b) in (23). To this end, we propose to adapt the phase shifts of the PS network to cancel the resulting inter-subarray phase difference. As term (b) corresponds to $-2\pi f_c \Delta t_k^n$, we propose to adjust the values of the PS network to attain a compensation phase difference of $+2\pi f_c \Delta t_k^n$, such that the resulting PS network based inter-subarray phase difference is nullified. This can be equalized by adding a phase shift of $+2\pi f_c t_k^n$ per subarray such that the phased array steering vector of the $k^{\text{th}}$ subarray belonging to the $n^{\text{th}}$ RF chain $\mathbf{a}_{n,k}$ is given as

$$\mathbf{a}_{n,k} = \bar{\mathbf{a}}_{n,k}e^{+j2\pi f_c t_k^n} \quad \forall\ k \in 1,2,.,K. \tag{24}$$

$\bar{\mathbf{a}}_{n,k}$ denotes the phased array steering vector of the $k^{\text{th}}$ subarray belonging to the $n^{\text{th}}$ RF chain, which is designed for the carrier frequency as follows

$$\bar{\mathbf{a}}_{n,k} = \frac{1}{\sqrt{N_T}}[e^{j(k-1)Q\Theta_c^{l,u}}, \ldots, e^{j(kQ-1)\Theta_c^{l,u}}]^T \;\forall\; k \in 1, 2, ., K. \tag{25}$$

Consequently, the PS network for $N_{\text{RF}}$ RF chains $\mathbf{A} \in \mathbb{C}^{N_T \times KN_{\text{RF}}}$ is modeled as

$$\mathbf{A} = [\mathbf{blkdiag}(\mathbf{a}_{1,1}, .., \mathbf{a}_{1,K}), .., \mathbf{blkdiag}(\mathbf{a}_{N_{\text{RF}},1}, .., \mathbf{a}_{N_{\text{RF}},K})], \tag{26}$$

where $\mathbf{a}_n = [\mathbf{a}_{n,1}^T, \ldots, \mathbf{a}_{n,K}^T]^T \in \mathbb{C}^{N_{\text{T}} \times 1}$ represents the phased array steering vector of the $n^{\text{th}}$ RF chain. The corresponding TTD network $\mathbf{X}_m \in \mathbb{C}^{KN_{\text{RF}} \times N_{\text{RF}}}$ at the $m^{\text{th}}$ subcarrier is modeled as

$$\mathbf{X}_m = \mathbf{blkdiag}(\mathbf{x}_m^1, .., \mathbf{x}_m^{N_{\text{RF}}}). \tag{27}$$

Based on the proposed hybrid analog precoder, the received symbol vector $\mathbf{y}_m \in \mathbb{C}^{N_{\text{s}} \times 1}$ at the $m^{\text{th}}$ subcarrier is given by

$$\mathbf{y}_m = \mathbf{H}_m \mathbf{A} \mathbf{X}_m \mathbf{D}_m \mathbf{s}_m + \mathbf{n}_m, \tag{28}$$

where $\mathbf{s}_m \in \mathbb{C}^{N_{\text{s}} \times 1}$ represents the input symbol vector, and $N_{\text{s}}$ denotes the number of data streams. $\mathbf{D}_m \in \mathbb{C}^{N_{\text{RF}} \times N_s}$ indicates the digital precoder at the $m^{\text{th}}$ subcarrier, and $\mathbf{n}_m \in \mathbb{C}^{U \times 1}$ denotes the complex additive white Gaussian noise (AWGN) vector with zero mean and $\sigma^2_{\text{noise}}$ variance.

In conclusion, the design of the hybrid precoding scheme proposed in this work is summarized in (19) and (24), wherein the elements of $\mathbf{t}_n^{\triangle}$ are upperbounded by $t_{\max}$. Thanks to the DDPP architecture, the proposed precoding scheme effectively reduces the maximum time delay required from TTDs by a factor of $K-1$. In consequence, the proposed architecture offers high degree of practicality for UM-MIMO systems, where a large number of TTDs is needed, as the severity of beam squint effect is directly proportional to the number of antenna elements [38].

### *B. True Time Delay Resolution Constraint*

The resolution of TTDs is an additional hardware limitation on the hybrid precoder, where time delay networks are commonly realized by time delay elements connected through switchable delay lines. This implementation imposes a finite resolution constraint on the time delay values realized by these networks. Therefore and aiming at a practical system design, the resolution constraint of TTDs should be taken into account in the design of the hybrid analog precoder. In this subsection, we propose a hybrid analog precoding scheme to address this limitation.

We analyze the time resolution error that occurs due to the adoption of finite-resolution TTDs considering the time delay vector $\mathbf{t}_n$ at the output of the TTD network. The time resolution error results from rounding the required time delay value to an integer multiple of the time delay resolution $t_{\text{res}}$. The time resolution error of the time delay value over the $k^{\text{th}}$ subarray belonging to the $n^{\text{th}}$ RF chain is given as

$$\tau_k^n = t_k^n - \lfloor \frac{t_k^n}{t_{\text{res}}} \rceil t_{\text{res}}. \tag{29}$$

After considering the time delay resolution constraint, the time delay vector $\mathbf{t}_n$ is expressed as

$$\mathbf{t}_n = \left[ \lfloor \frac{t_1^n}{t_{\text{res}}} \rceil t_{\text{res}}, \lfloor \frac{t_2^n}{t_{\text{res}}} \rceil t_{\text{res}}, \ldots, \lfloor \frac{t_K^n}{t_{\text{res}}} \rceil t_{\text{res}} \right]^T. \tag{30}$$

Since a TTD element delivers frequency-dependent phase shifts, it also produces frequency-dependent phase errors due to the time resolution error. Therefore, we propose to convert the time error resulted from the time delay resolution into a phase error, which we compensate for by introducing an equalization phase shift to the PS network. We can model the time resolution error $\tau_k^n$ as a vector of frequency-dependent phase errors $\boldsymbol{\varphi}_k^n \in \mathbb{R}^{M \times 1}$ expressed as

$$\boldsymbol{\varphi}_k^n = -2\pi \mathbf{f} \tau_k^n = \begin{pmatrix} \varphi_{k,1}^n \\ \cdot \\ \cdot \\ \varphi_{k,M}^n \end{pmatrix}, \tag{31}$$

where $\varphi_{k,m}^n$ denotes the phase error resulted at the $m^{\text{th}}$ subcarrier due to the time delay resolution. We propose to compensate these phase shift errors by adding equalization phase shifts $\phi_k^n$ to the PS network. We model this optimization problem as the minimization of the squared frobenius norm of the sum of the phase errors $\boldsymbol{\varphi}_k^n$ and the equalization phase $\phi_k^n$ for a given subarray. Accordingly, the problem is formulated as

$$\text{P1:} \min_{\phi_k^n} s \| -2\pi \mathbf{f} \tau_k^n + \phi_k^n \|_F^2. \tag{32}$$

Let's define $\phi_k^n = 2\pi \rho_k^n \tau_k^n$, where $\rho_k^n$ is an auxilliary optimization variable corresponding to the equalization phase $\phi_k^n$. Accordingly, Problem P1 can be rewritten as

$$\text{P2:} \min_{\rho_k^n} \left\| -2\pi \begin{pmatrix} f_1 - \rho_k^n \\ \cdot \\ \cdot \\ f_M - \rho_k^n \end{pmatrix} \tau_k^n \right\|_F^2, \tag{33}$$

where Problem P2 can be simplified to

$$\text{P3:} \min_{\rho_k^n} \sum_{m=1}^{M} (\rho_k^n - f_m)^2. \tag{34}$$

By substituting $f_m = f_c + \frac{B}{M}[m - \frac{M+1}{2}]$, the objective function of Problem P3 can be expressed as

$$\sum_{m=1}^{M} \left[ \rho_k^n - f_c - \frac{B}{M} \left( m - \frac{M+1}{2} \right) \right]^2. \tag{35}$$

Following linear algebra, Problem P3 can be rewritten as

$$\text{P4:} \min_{\rho_k^n} \Bigg[ M \rho_k^{n\,2} - 2M f_c \rho_k^n + f_c^2 M + \left( \frac{B}{M} \right)^2 \left( \frac{-2M^3 - 3M^2 + 6M + 3}{12} \right) \Bigg]. \tag{36}$$

We obtain the value of $\rho_k^n$ that solves Problem P4 by setting the derivative of the objective function in P4 with respect to $\rho_k^n$ to zero. This can be expressed as follows

$$2M\rho_k^n + -2Mf_c = 0. \tag{37}$$

As a result of (37), $\rho_k^n = f_c$, and, accordingly, the corresponding equalization phase is $\phi_k^n = 2\pi f_c \tau_k^n$. The equalization phase $\phi_k^n$ is added to the corresponding subarray, resulting in an adjusted phased array steering vector as

$$\mathbf{a}_{n,k} = \bar{\mathbf{a}}_{n,k}.e^{+j2\pi f_c[t_k^n+\tau_k^n]} \quad \forall\ k \in 1, 2, .., K. \tag{38}$$

We determine the time delay vector $\mathbf{t}_n^\triangle$ for the DDPP architecture by substituting (30) in (18). The elements of $\mathbf{t}_n^\triangle$ are then upperbounded by $t_{\max}$ to comply with the maximum delay range constraint of TTDs. The switch closes to its first output if $\mathbf{t}_n^\triangle \succeq \mathbf{0}_{K-1,1}$, otherwise it closes to the second output. We denote this precoding scheme as the time resolution constrained hybrid precoding (TR-HP).

The proposed DDPP based TR-HP effectively addresses the hardware limitations of TTDs, enabling practical implementation of wideband beamforming under TTD contraints.

## IV. Hybrid Analog Precoding Under True Time Delay and Phase Shifter Constraints

In the preceding section, we propose hybrid precoding schemes that incorporate practical finite-resolution and limited-range TTDs with infinite-resolution PSs. In the same vein, recent works in the literature have also used infinite-resolution PSs in the development of hybrid analog precoders [25], [33], [40], which is impractical due to the associated cost and power consumption in UM-MIMO systems. Therefore, in this part, we propose a hybrid analog precoding scheme that accounts for both TTD and PS hardware limitations, which we denote as the hybrid practical precoding (HPP). The problem of interest is to design the elements of $\mathbf{t}_n$ and $\mathbf{A}$ in order to minimize the Euclidean distance between the optimal analog precoder $\mathbf{C}_m$ and the resulting hybrid precoder over all subcarriers subjected to low-resolution TTDs and PSs constraints. We hereby confine the optimization domain of $\mathbf{t}_n$ to a set of discrete time delay values $\Upsilon$, where the elements of $\Upsilon$ are integers multiple of the time delay resolution $t_{\text{res}}$. Similarly, non-zero elements of $\mathbf{A}$ belong to a set $\Gamma$, which contains quantized phase shifts with constant modulus $\frac{1}{\sqrt{N_\text{T}}}$. Accordingly, the problem is formulated as

$$\text{P5:}\ \underset{\mathbf{A},\{\mathbf{t}_n\}_{n=1}^{N_{RF}}}{\text{minimize}} \sum_{m=1}^{M} \left\| \mathbf{C}_m - \mathbf{A}\mathbf{X}_m(\{\mathbf{t}_n\}_{n=1}^{N_{RF}}) \right\|_F^2, \tag{39a}$$
$$\text{s.t.}\ \mathbf{A}(w,r) \in \{0, \Gamma\}\ \forall\ w \in 1, 2, ., N_\text{T},\ r \in 1, 2, ., KN_\text{RF}, \tag{39b}$$
$$t_k^n \in \Upsilon\ \forall\ n \in 1, 2, .., KN_\text{RF},\ k \in 1, 2, ., K. \tag{39c}$$

The constraint (39b) indicates the sparsity of $\mathbf{A}$, whose elements are either zeros or belong to $\Gamma$. The constraint (39c) denotes the time delay resolution imposed on the elements of $\mathbf{t}_n$.

The objective function of Problem P5 in (39a) is equivalently written as

$$\sum_{m=1}^{M}\sum_{n=1}^{N_\text{RF}}\sum_{k=1}^{K} \left\| \mathbf{c}_{n,k,m} - \mathbf{a}_{n,k} e^{-j2\pi f_m t_k^n} \right\|_F^2, \tag{40}$$

where $\mathbf{c}_{n,k,m} \in \mathbb{C}^{Q\times 1}$ is a subvector of the $n^{\text{th}}$ column of $\mathbf{C}_m$ corresponding to the $k^{\text{th}}$ subarray.

By introducing $\mathbf{b}(t_k^n) = e^{-j2\pi \mathbf{f} t_k^n} \in \mathbb{C}^{M\times 1}$ as the phase shifts' vector resulting from the time delay value $t_k^n$ over $M$ subcarriers, the objective function in (40) can be rewritten as

$$\sum_{n=1}^{N_\text{RF}}\sum_{k=1}^{K} \left\| \mathbf{C}_{n,k} - \mathbf{a}_{n,k}\mathbf{b}^T(t_k^n) \right\|_F^2, \tag{41}$$

where $\mathbf{C}_{n,k} = [\mathbf{c}_{n,k,1}, \ldots, \mathbf{c}_{n,k,M}] \in \mathbb{C}^{Q\times M}$.

From (41), we can conclude that our problem in P5 is equivalent to the minimization of the contribution of each subarray and its time delay value to the overall objective function. Hence, Problem P5 can be divided into $KN_\text{RF}$ subproblems, where each subproblem handles the joint optimization between the time delay value and phase shifts of a given subarray. We formulate the joint optimization subproblem of the $k^{\text{th}}$ subarray belonging to the $n^{\text{th}}$ RF chain subjected to the time delay and phase shift resolutions as

$$\text{P6:}\ \underset{\mathbf{a}_{n,k},t_k^n}{\text{minimize}} \left\| \mathbf{C}_{n,k} - \mathbf{a}_{n,k}\mathbf{b}^T(t_k^n) \right\|_F^2, \tag{42a}$$
$$\text{s.t.}\ \mathbf{a}_{n,k}(q) \in \Gamma \quad \forall\ q = 1, 2, ., Q, \tag{42b}$$
$$t_k^n \in \Upsilon. \tag{42c}$$

Problem P6 relieves the sparsity limitation imposed on the optimization variables of Problem P5, where $\mathbf{a}_{n,k}$ and $\mathbf{b}(t_k^n)$ are non-sparse variables. However, this joint optimization problem is intractable, where non-convex constraints (42b) and (42c) hinder finding a global optimal solution. Specifically, the constraint (42c) stipulates that the obtained time delay must belong to a set of discrete time delay values, denoted as $\Upsilon$. Because the optimization variable is constrained to be discrete, the problem becomes a mixed-integer problem, which is well-known to be NP-hard. Therefore, we propose next an iterative optimization using alternating minimization in order to handle such NP-hard problem [41].

Alternating minimization is a common practice to solve such non-convex optimization problems with several types of variables. Therefore, we propose an alternating minimization based iterative algorithm to find a local optimal solution to Problem P6. Specifically, the joint optimization problem in P6 is decoupled into two problems. The first problem P7 optimizes PS precoding vector $\mathbf{a}_{n,k}$ while $t_k^n$ is fixed, which is formulated as

$$\text{P7:}\ \underset{\mathbf{a}_{n,k}}{\text{minimize}} \left\| \mathbf{C}_{n,k} - \mathbf{a}_{n,k}\mathbf{b}^T(t_k^n) \right\|_F^2, \tag{43a}$$
$$\text{s.t.}\ \mathbf{a}_{n,k}(q) \in \Gamma \quad \forall\ q = 1, 2, .., Q. \tag{43b}$$

The second problem P8 optimizes $t_k^n$ while $\mathbf{a}_{n,k}$ is fixed, which is formulated as

$$\text{P8:}\ \underset{t_k^n}{\text{minimize}} \left\| \mathbf{C}_{n,k} - \mathbf{a}_{n,k}\mathbf{b}^T(t_k^n) \right\|_F^2, \tag{44a}$$

**Algorithm 1** Finding Time Delay Value

**Input:** $\mathbf{b}(t_k^n)$, $\mathbf{f}$
**Output:** $t_k^n$
**Initialization:** $i=0$, $\Omega=10^6$
**for** $p=-\xi\ \ldots\ \xi$ **do**
  $i=i+1$
  $t_i = \mathrm{Re}\left\{\dfrac{1}{-j2\pi\sum_{m=1}^{M} f_m^2}\mathbf{f}^*[\mathrm{Ln}[\mathbf{b}(t_k^n)]+jp2\pi]\right\}$
  **if** $(\|e^{-j2\pi\mathbf{f}t_i}-\mathbf{b}(t_k^n)\|_F^2<\Omega)$ **then**
    $\Omega = \|\, e^{-j2\pi\mathbf{f}t_i}-\mathbf{b}(t_k^n)\|_F^2$
    **update** $t_k^n \leftarrow t_i$
  **end if**
**end for**
**return** $t_k^n$

$$\text{s.t.}\ \ t_k^n \in \Upsilon. \tag{44b}$$

By temporarily neglecting the constraint (43b), Problem P7 has an optimal solution given as

$$\mathbf{a}_{n,k} = \frac{1}{M}\mathbf{C}_{n,k}\mathbf{b}^*(t_k^n). \tag{45}$$

Similarly, by temporarily neglecting the constraint (44b), Problem P8 has an optimal solution for $\mathbf{b}(t_k^n)$ given as

$$\mathbf{b}(t_k^n) = \frac{N_\mathrm{T}}{Q}(\mathbf{a}_{n,k}^H\mathbf{C}_{n,k})^T. \tag{46}$$

However, we still need to derive the time delay value $t_k^n$ corresponding to $\mathbf{b}(t_k^n)$. Knowing that $\mathbf{b}(t_k^n) = e^{-j2\pi\mathbf{f}t_k^n}$, we can write

$$\ln[\mathbf{b}(t_k^n)] = -j2\pi\mathbf{f}t_k^n. \tag{47}$$

Consequently, time delay value $t_k^n$ can be isolated as [42]

$$t_k^n = \mathrm{Re}\left\{\frac{1}{-j2\pi\sum_{m=1}^{M} f_m^2}\mathbf{f}^*\{\mathrm{Ln}[\mathbf{b}(t_k^n)]+jp2\pi\}\right\}, \tag{48}$$

where $p\in\mathbb{Z}$. Since the complex exponential function is periodic with a period of $j2\pi$, its complex logarithm is described as multi-valued function. Particularly, (48) holds as a solution for any imaginary number $j2\pi p$ added to its logarithmic expression. Therefore, we propose to perform a one-dimensional search over a probing integer $p\in[-\xi,\xi]$ to find the time value $t_k^n$ that solves Problem P9, which is formulated as

$$\text{P9:}\ \underset{t_k^n}{\text{minimize}} \left\|e^{-j2\pi\mathbf{f}t_k^n}-\mathbf{b}(t_k^n)\right\|_F^2. \tag{49}$$

The value of $p$ corresponding to the searched time value $t_k^n$ is directly proportional to the subcarrier frequency and maximum required time delay over subarrays. Therefore, a search parameter is recommended to be $\xi = f_M t_\mathrm{req}$. The proposed procedure is summarized in Algorithm 1.

Now, we propose a joint iterative optimization algorithm to solve problems in P7 and P8 alternately with the consideration of constraints (43b) and (44b). The iterative optimization starts

**Algorithm 2** Hybrid Practical Precoding (HPP)

**Input:** $\mathbf{f}$, $\mathbf{C}_n^\star$, $\Upsilon$, $N_\mathrm{T}$, $Q$, $M$, $Iter, \epsilon$
**Output:** $\mathbf{a}_n, \mathbf{t}_n, \mathbf{t}_n^\triangle$
**Initialization:** $\mathbf{a}_n = \mathbf{0}_{N_\mathrm{T},1}, \mathbf{t}_n = \mathbf{0}_{K,1}, \mathbf{t}_n^\triangle = \mathbf{0}_{K-1,1}$
**Parallel for** $k=1:K$ **do**
  $i=0$
  $\mathbf{F}_{n,k}^0 = \mathbf{0}_{Q,M}$
  $\tilde{\mathbf{a}}_{n,k} = \dfrac{1}{\sqrt{N_\mathrm{T}}}\mathbf{1}_Q$
  $\mathbf{C}_{n,k} = \mathbf{C}_n^\star((k-1)Q+1:kQ,:)$
  **while** *true* **do**
    $i=i+1$
    **update** $\mathbf{b}(t_k^n) \leftarrow \dfrac{N_\mathrm{T}}{Q}(\tilde{\mathbf{a}}_{n,k}^H\mathbf{C}_{n,k})^T$
    $t_k^n = \mathrm{Algorithm1}[\mathbf{b}(t_k^n)]$
    $\tilde{t}_k^n = \mathbf{QZ}_\Upsilon(t_k^n)$
    **update** $\mathbf{a}_{n,k} \leftarrow \dfrac{1}{M}\mathbf{C}_{n,k}\mathbf{b}^*(\tilde{t}_k^n)$
    $\tilde{\mathbf{a}}_{n,k} = \mathbf{QZ}_\Gamma(\mathbf{a}_{n,k})$
    $\mathbf{F}_{n,k}^i = \tilde{\mathbf{a}}_{n,k}\mathbf{b}^T(\tilde{t}_k^n)$
    **if** $(\left\|\mathbf{F}_{n,k}^i - \mathbf{F}_{n,k}^{i-1}\right\|_F^2 < \epsilon \parallel i = Iter)$ **then**
      $\mathbf{a}_n = [\mathbf{a}_n \mid \tilde{\mathbf{a}}_{n,k}]$
      $\mathbf{t}_n = [\mathbf{t}_n \mid \tilde{t}_k^n]$
      **break**
  **end while**
**end for**
**Parallel for** $k=1:K-1$ **do**
  $\Delta t_k^n = \mathrm{sign}(\tilde{t}_{k+1}^n - \tilde{t}_k^n)\min(|\tilde{t}_{k+1}^n - \tilde{t}_k^n|, t_{max})$
  $\mathbf{t}_n^\triangle = [\mathbf{t}_n^\triangle \mid \Delta t_k^n]$
**end for**
**return** $\mathbf{a}_n, \mathbf{t}_n, \mathbf{t}_n^\triangle$

by initializing the phased array vector $\mathbf{a}_{n,k}$ as an all-ones vector scaled by a constant modulus $1/\sqrt{N_\mathrm{T}}$. Based on $\mathbf{a}_{n,k}$, the phase shifts vector over subcarriers $\mathbf{b}(t_k^n)$ is calculated using (46). Afterwards, the resulting vector is passed to Algorithm 1 to find the corresponding time delay value $t_k^n$. Subsequently, the resulting time value is quantized to the nearest element in $\Upsilon$ to satisfy (44b). Based on the resulting time value $\tilde{t}_k^n$, the phased array vector $\mathbf{a}_{n,k}$ is updated using (45) and the elements of the resulting vector are quantized to the nearest phase shifts in $\Gamma$ to satisfy (43b). These steps are iteratively performed until convergence. The convergence is met once the Euclidean distance between the actual hybrid precoder per subarray $\mathbf{F}_{n,k}^i$ and that resulted in the previous iteration $\mathbf{F}_{n,k}^{i-1}$ is lower than a threshold $\epsilon$ or by reaching the maximum number of iterations, i.e. $Iter$. The time delay vector of TTDs for the DDPP architecture $\mathbf{t}_n^\triangle$ is determined using (18). The maximum delay range constraint is imposed on the elements of $\mathbf{t}_n^\triangle$, where the elements of $\mathbf{t}_n^\triangle$ are upperbounded by $t_\mathrm{max}$. The switch closes to the first output if $\mathbf{t}_n^\triangle \succeq \mathbf{0}_{K-1,1}$, while it closes to the second output otherwise. The proposed algorithm is executed in parallel on subarrays' level of a specific RF chain and scaled up to multiple RF chains. The proposed algorithm is summarized in Algorithm 2 where $\mathbf{C}_n^\star = [\mathbf{C}_{n,1}^T, \ldots, \mathbf{C}_{n,K}^T]^T \in \mathbb{C}^{N_\mathrm{T} \times M}$ denotes the full-TTD precoder of subarrays belonging to the $n^{\text{th}}$ RF chain.

## V. Fixed-Delta Hybrid Practical Precoding

The proposed hybrid practical precoding scheme addresses the resolution constraints of TTDs and PSs. However, it does not ensure identical inter-subarray time delay differences across all subarrays, which is desirable for practical cost-efficient design [37]. Therefore, we propose customizing HPP scheme to ensure both identical inter-subarray time delay differences and reduced computational complexity. We denote this precoding scheme as the fixed-delta hybrid practical precoding (FD-HPP). We confine Problem P6 to the first pair of subarrays as follows

$$\text{P10:} \ \underset{\mathbf{a}_{n,k}, t_k^n}{\text{minimize}} \left\| \mathbf{C}_{n,k} - \mathbf{a}_{n,k} \mathbf{b}^T(t_k^n) \right\|_F^2, \tag{50a}$$

$$\text{s.t.} \ \ \mathbf{a}_{n,k}(q) \in \Gamma \quad \forall\ q = 1, 2, .., Q,\ k = 1, 2, \tag{50b}$$

$$t_k^n \in \Upsilon \quad \forall\ k = 1, 2. \tag{50c}$$

We derive the time delay values for the first two subarrays by solving Problem P10 in the same iterative manner as for Problem P6 in (42a). Based on the derived time delay values $t_1^n$ and $t_2^n$, we calculate the inter-subarray time delay value between the first pair of subarrays as

$$\Delta t_1^n = t_2^n - t_1^n. \tag{51}$$

We adopt this inter-subarray time delay value for all successive TTDs such that the time delay vector of TTDs belonging to $n^{\text{th}}$ RF chain is expressed as

$$\mathbf{t}_n^\triangle = \Delta t_1^n \mathbf{1}_{K-1}. \tag{52}$$

The switch closes to the first output if $\Delta t_1^n \geq 0$, while it closes to the second output otherwise. The corresponding phase shifts are calculated using (45) and subsequently quantized to the nearest elements in $\Gamma$. The proposed algorithm, which is summarized in Algorithm 3, guarantees identical inter-subarray time delay differences while maintaining lower computational overhead.

The computational complexity of HPP is dominated by steps 9, 10, 12, and 14. In specific, the complexity of steps 9, 12, and 14 is $\mathcal{O}(M \cdot Q)$ whereas the complexity of step 10, i.e. Algorithm 1, is $\mathcal{O}(\xi \cdot M)$. As HPP runs for $Iter$ iterations per subarray, it has a complexity of $[\mathcal{O}(K \cdot Iter \cdot M \cdot Q) + \mathcal{O}(K \cdot Iter \cdot \xi \cdot M)]$. On the other hand, the proposed FD-HPP features lower computational overhead with a factor of $K$ where the iterative loop runs only for the first two subarrays. Therefore, it has a complexity of $[\mathcal{O}(Iter \cdot M \cdot Q) + \mathcal{O}(Iter \cdot \xi \cdot M)] < [\mathcal{O}(K \cdot Iter \cdot M \cdot Q) + \mathcal{O}(K \cdot Iter \cdot \xi \cdot M)]$.

## VI. Power and Computational Metrics of the Proposed Schemes

In this section, we first evaluate the reduction in power consumption achieved by using practical, limited-range TTDs in our proposed DDPP architecture, as compared to the conventional DPP architecture. We then delve into the computational complexity of the proposed iterative schemes–specifically,

**Algorithm 3** Fixed-Delta Hybrid Practical Precoding (FD-HPP)

**Input:** $\mathbf{f},\ \mathbf{C}_n^\star,\ t_{\max},\ N_T,\ Q,\ M,\ Iter, \epsilon$
**Output:** $\mathbf{a}_n, \mathbf{t}_n, \mathbf{t}_n^\triangle$

**Initialization:**
$\mathbf{a}_n = \mathbf{0}_{N_T,1}, \mathbf{t}_n = \mathbf{0}_{K,1}, \mathbf{t}_n^\triangle = \mathbf{0}_{K-1,1}$
**Parallel for** $k = 1:2$ **do**
  Perform steps (3)-(21) of Algorithm 2
**end for**
$\Delta t_1^n = \tilde{t}_2^n - \tilde{t}_1^n$
$\mathbf{t}_n^\triangle = \Delta t_1^n \mathbf{1}_{K-1}$
$\mathbf{t}_n = [0, 1, \ldots, K-1]^T \Delta t_1^n$
**Parallel for** $k = 3:K$ **do**
  $\mathbf{C}_{n,k} = \mathbf{C}_n^\star((k-1)Q + 1 : kQ, :)$
  $\tilde{\mathbf{a}}_{n,k} = \mathbf{Q}\mathbf{Z}_\Gamma[\frac{1}{M}\mathbf{C}_{n,k} e^{j2\pi \mathbf{f} t_k^n}]$
  $\mathbf{a}_n = [\mathbf{a}_n \mid \tilde{\mathbf{a}}_{n,k}]$
**end for**
**return** $\mathbf{a}_n, \mathbf{t}_n, \mathbf{t}_n^\triangle$

HPP and FD-HPP schemes. These are designed to operate in a practical system with finite-resolution TTDs and PSs.

### A. Power Consumption of the DDPP Architecture

In this subsection, we evaluate the power consumption of the proposed DDPP architecture compared to the traditional DPP architecture. The power consumption of the DPP architecture is expressed as

$$P_{\mathrm{DPP}} = \chi + K N_{\mathrm{TTD}}^{\mathrm{DPP}} P_{\mathrm{TTD}}, \tag{53}$$

where $\chi = N_{\mathrm{RF}} P_{\mathrm{RF}} + N_{\mathrm{RF}} N_{\mathrm{T}} P_{\mathrm{PS}} + N_{\mathrm{RF}} K P_{\mathrm{PD}} + P_{\mathrm{PC}} N_{\mathrm{T}}$ such that $P_{\mathrm{RF}}$, $P_{\mathrm{TTD}}$, $P_{\mathrm{PS}}$, $P_{\mathrm{PD}}$, $P_{\mathrm{PC}}$ denote the power consumption of the RF chain, a single TTD, the phase shifter, the power divider, and the power combiner, respectively. $N_{\mathrm{TTD}}^{\mathrm{DPP}}$ represents the required number of cascaded TTDs per subarray in the DPP architecture. Similarly, the power consumption of the proposed DDPP architecture is expressed as

$$P_{\mathrm{DDPP}} = \chi + K N_{\mathrm{TTD}}^{\mathrm{DDPP}} P_{\mathrm{TTD}} + N_{\mathrm{RF}} P_{\mathrm{SW}}, \tag{54}$$

where $P_{\mathrm{SW}}$ and $N_{\mathrm{TTD}}^{\mathrm{DDPP}}$ denote the power consumption of a 2-output switch and the number of cascaded TTDs per subarray in the DDPP architecture, respectively. Upon comparing Equations (53) and (54), it becomes apparent that the primary difference in power consumption between the DPP and DDPP architectures lies in the TTD network. Specifically, the number of switches is constrained by the limited number of RF chains $N_{\mathrm{RF}}$, and the power consumption of the switch is minimal compared to other components of the hybrid beamformer [39].

We consider the TTD chip proposed in [28], which operates at THz band and achieves a maximum delay range $t_{\max} = 6.64$ ps with a time resolution of 0.44 ps. This chip can be reconfigured to realize an enhanced maximum delay range of 14.272 ps, but with a coarser time resolution of 3.568 ps. By cascading multiple TTDs, the maximum delay range of a single TTD can be expanded as shown in [32]. In the case of the DPP architecture, which requires a time delay range up to 400 ps with a fine resolution when $K = 16$ and $N_{\mathrm{T}} = 256$, nearly $N_{\mathrm{TTD}}^{\mathrm{DPP}} = 60$ TTDs would need to be cascaded per subarray. In contrast, the proposed DDPP architecture demands a time delay range of only 28 ps with a resolution of 4 ps for the same $K$ and $N_{\mathrm{T}}$. Thus, the DDPP architecture necessitates merely $N_{\mathrm{TTD}}^{\mathrm{DDPP}} = 2$ cascading TTDs per subarray. Consequently, the proposed DDPP architecture achieves lower power consumption than the DPP architecture, which can be attributed to the adoption of practical limited-range TTDs.

### *B. Computational Complexity of the HPP and FD-HPP Schemes*

In this subsection, we analyze the computational complexity of the proposed HPP and FD-HPP schemes in terms of fractional bandwidth and the number of antenna elements.

The computational complexity of the proposed HPP scheme is given as $[\mathcal{O}(K \cdot Iter \cdot M \cdot Q) + \mathcal{O}(K \cdot Iter \cdot \xi \cdot M)]$. It can be observed that the computational complexity of the HPP scheme increases linearly with both the search parameter $\xi$ and the number of TTDs $K$. The search parameter $\xi$ is given by $\xi = f_M A_{\mathrm{phys}}/v$. It's important to note that $\xi$ increases linearly with $f_M$. As a result, the search parameter $\xi$ increases linearly with the fractional bandwidth since $f_M$ is proportional to the bandwidth. Similarly, the required number of TTDs increases linearly with the fractional bandwidth [25]. Hence, we conclude that the computational complexity of the proposed HPP scheme increases quadratically with the fractional bandwidth $\beta$. In terms of antenna-related complexity, the first term $\mathcal{O}(K \cdot Iter \cdot M \cdot Q)$ increases linearly with the number of antennas since $KQ = KN_{\mathrm{T}}/K = N_{\mathrm{T}}$. Similarly, the complexity of the second term $\mathcal{O}(K \cdot Iter \cdot \xi \cdot M)$ increases linearly with the number of antennas since the search parameter $\xi$ has a linear relationship with the array aperture $A_{\mathrm{phys}}$. Consequently, the computational complexity of the proposed HPP scheme increases linearly with the number of antennas $N_{\mathrm{T}}$.

In order to reduce the computational complexity, we proposed the FD-HPP scheme, whose computational complexity is given as $[\mathcal{O}(Iter \cdot M \cdot Q) + \mathcal{O}(Iter \cdot \xi \cdot M)]$. We clearly observe that the computational complexity of the FD-HPP scheme is independent on the number of TTDs $K$. Specifically, the computational complexity increases linearly only with $\xi$; hence, the computational complexity of the FD-HPP scheme increases linearly with the fractional bandwidth $\beta$. In terms of the complexity concerning the number of antennas, the computational complexity of the first term $\mathcal{O}(Iter \cdot M \cdot Q)$ remains unchanged with an increase in the number of antennas, provided the number of antennas per subarray $Q$ remains constant. On the other hand, the computational complexity of the second term $\mathcal{O}(Iter \cdot \xi \cdot M)$ grows linearly with the search parameter $\xi$, thus linearly with the number of antennas. Accordingly, we conclude that the computational complexity of the FD-HPP scheme swells linearly with the number of antennas $N_{\mathrm{T}}$.

In summary, the computational complexity of the proposed HPP scheme grows quadratically with the fractional bandwidth and linearly with the number of antennas. In contrast, the FD-HPP scheme's complexity rises merely linearly with both the fractional bandwidth and the number of antennas, positioning the FD-HPP scheme as a more computationally efficient alternative to the HPP scheme.

TABLE I
SIMULATION PARAMETERS

| | |
|---|---|
| **Carrier frequency ($f_c$)** | 300 GHz |
| **Bandwidth ($B$)** | 30 GHz |
| **Number of subcarriers ($M$)** | 128 |
| **Number of antennas ($N_{\mathrm{T}}$)** | 256 |
| **Interelement distance ($d$)** | $\lambda_c/2$ |
| **Number of TTDs ($K$)** | 16 |
| **Number of RF chains ($N_{\mathrm{RF}}$)** | 4 |
| **Number of paths per user ($L$)** | 1 |
| **Maximum delay range ($t_{\max}$)** | 28, 340 ps |
| **Time delay resolution ($t_{\mathrm{res}}$)** | 4ps |
| **Number of bits controlling the PS ($b$)** | 2 bits |
| **Number of iterations ($Iter$)** | 5 |
| **Threshold ($\epsilon$)** | $10^{-2}$ |

## VII. SIMULATION RESULTS

In this section, we present a performance evaluation of the proposed DDPP architecture and the precoding schemes. To ensure a comprehensive analysis, we compare our work to the state-of-the-art techniques presented in references [25] and [33]. We consider a wideband UM-MIMO-OFDM system with $M = 128$ subcarriers operating at a carrier frequency of $f_c = 300$ GHz and a bandwidth of $B = 30$ GHz. The BS in this system is equipped with a ULA consisting of $N_{\mathrm{T}} = 256$ half-wavelength spaced antennas. The BS also incorporates $K = 16$ TTDs, along with $N_{\mathrm{RF}} = 4$ RF chains. The THz band channel is known for its highly directional propagation, and is commonly referred to as a LoS-dominant channel [43], [44]. As such, we assume that there is only one path per user, i.e., $L = 1$. With regard to hardware constraints, the proposed DDPP architecture requires a maximum time delay of 26.67 ps for $\theta_{l,u} = \pi/2$, as calculated using (17). Therefore, we restrict the TTDs to a maximum delay range of $t_{\max} = 28$ ps. This maximum delay range along with a time resolution of almost $t_{\mathrm{res}} = 4$ ps can be realized by the cascading of only two TTDs chips, as elaborated in subsection VI-A. For the sake of comparison, we also consider TTDs with a maximum delay range of $t_{\max} = 340$ ps as used in [33]. Besides, we consider low-resolution PSs digitally controlled by $b = 2$ bits. We assume $Iter = 5$ and $\epsilon = 10^{-2}$ as the convergence conditions for the proposed HPP and FD-HPP schemes. The simulation parameters are summarized in TABLE I.

In terms of evaluation metrics, we consider the normalized array gain per user $g(\theta_{l,u})$, which is defined as the array gain normalized to that achieved by an unconstrained full-TTD precoder averaged over subcarriers for a given user and expressed as

$$g(\theta_{l,u}) = \frac{1}{M}\sum_{m=1}^{M} g_m(\theta_{l,u}). \tag{55}$$

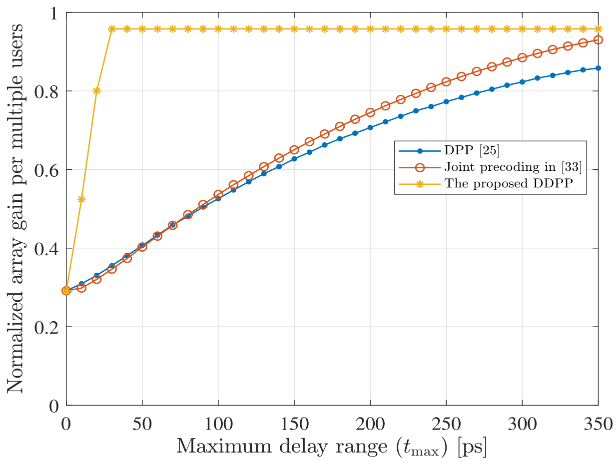


Fig. 4. Normalized array gain per multiple users versus maximum delay range $t_{\max}$.

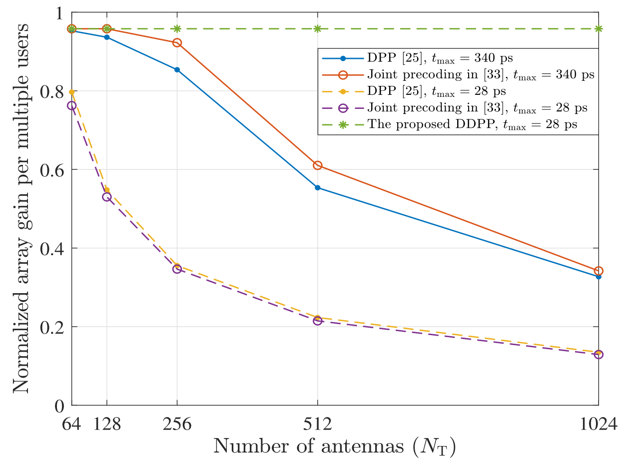


Fig. 5. Normalized array gain per multiple users versus number of antennas $N_{\mathrm{T}}$ when $t_{\max} = 28$ and 340 ps.

Furthermore, we also consider the normalized array gain per multiple users with random orientations $\theta_{l,u} \in [0, \pi]$, which is given as

$$\bar{g} = \frac{1}{U} \sum_{u=1}^{U} g(\theta_{l,u}). \tag{56}$$

Additionally, we consider the spectral efficiency per subcarrier, which is calculated as

$$R = \frac{1}{M} \sum_{m=1}^{M} \log_2 \left( \mathbf{I}_U + SNR.\mathbf{H}_{\mathrm{m}} \mathbf{A} \mathbf{X}_m \mathbf{D}_m \mathbf{D}_m^H \mathbf{X}_m^H \mathbf{A}^H \mathbf{H}_{\mathrm{ml}}^H \right), \tag{57}$$

where $SNR$ denotes the signal-to-noise ratio per subcarrier. In order to suppress inter-user interference, we adopt zero-forcing digital precoding $\mathbf{D}_m = \mathbf{H}_{\mathrm{eff},m}^H (\mathbf{H}_{\mathrm{eff},m} \mathbf{H}_{\mathrm{eff},m}^H)^{-1}$, where $\mathbf{H}_{\mathrm{eff},m} = \mathbf{H}_m \mathbf{A} \mathbf{X}_m$ denotes the effective channel at the $m^{\mathrm{th}}$ subcarrier. The digital precoder $\mathbf{D}_m$ is normalized where the total transmit power constraint is met, such that $\|\mathbf{A}\mathbf{X}_m \mathbf{D}_m\|_F^2 = N_{\mathrm{s}}$. Simulation results are averaged over 1000 iterations.

### *A. Hybrid Analog Precoding Under Limited-Range TTD Constraint*

In this subsection, we present a performance comparison of the proposed DDPP precoding scheme with the hybrid precoding schemes proposed in [25] and [33] considering practical limited-range TTDs.

Fig. 4 presents the normalized array gain per multiple users versus the maximum delay range $t_{\max}$. It can be observed that the normalized array gain per multiple users increases as $t_{\max}$ scales up. For $t_{\max} = 10$ ps, the normalized array gain per multiple users of the hybrid precoding in [25] is nearly 0.3, which increases to 0.86 for $t_{\max} = 350$ ps. The hybrid precoding in [33] achieves a slightly higher performance, reaching 0.93 for $t_{\max} = 350$ ps. In contrast, the proposed DDPP achieves a much higher performance of 0.96 for $t_{\max} = 28$ ps. This remarkable improvement is due to the proposed DDPP architecture, which does not require the maximum time delay to scale up with the TTD's index. In particular, the maximum time delay required by the proposed DDPP is $\Delta t_k^n$, whereas the hybrid precoding in [25] requires $(K-1)\Delta t_k^n$. When $\theta_{l,u} = \pi/2$, the maximum time delay required by the proposed DDPP is 26.67 ps per (17), whereas a maximum time delay of 400 ps is required by the DPP architecture as calculated by (14).

Fig. 5 depicts the normalized array gain per multiple users versus the number of antennas $N_{\mathrm{T}}$ for $t_{\max} = 340$ ps and $t_{\max} = 28$ ps. We increase the number of TTDs $K \in \{4, 8, 16, 32, 64\}$ as the number of antenna elements increases $N_{\mathrm{T}} = \{64, 128, 256, 512, 1024\}$ such that the ratio $Q = N_{\mathrm{T}}/K = 16$ remains constant, thereby ensuring the same level of robustness against beam squint effect. The results show that the normalized array gain per multiple users of the hybrid precodings in [25] and [33] decreases significantly as the number of antennas increases. Specifically, it drops from nearly 0.8 to less than 0.2 for $t_{\max} = 28$ ps, and from 0.96 to less than 0.4 for $t_{\max} = 340$ ps. This degradation is due to the maximum required delay range of [25] and [33], which increases with the number of TTDs. In contrast, the proposed DDPP maintains a normalized array gain per multiple users of 0.96 as the number of antenna elements increases for $t_{\max} = 28$ ps. This is due to the maximum required time delay not scaling with $K$, making the proposed DDPP scalable and thus suitable for UM-MIMO systems.

With the aim of investigating the performance of the proposed DDPP precoding scheme deeper, we plot in Fig. 6 the normalized array gain over subcarriers for DDPP compared to the hybrid precoding schemes in [25] and [33]. We consider practical limited-range TTDs with $t_{\max} = 28$ ps and $t_{\max} = 340$ ps, where $N_{\mathrm{T}} = 256$, $K = 16$, and $\sin(\theta_{l,u}) = 0.8$. Our investigation reveals that the proposed DDPP exhibits the same degree of robustness against beam squint effect as the hybrid precoding in [33] when $t_{\max} = 340$ ps. Specifically, the normalized array gain of all subcarriers is higher than 0.8 in both approaches. Notably, the proposed DDPP achieves a higher degree of robustness against beam squint effect than DPP approach in [25], where the normalized array gain of

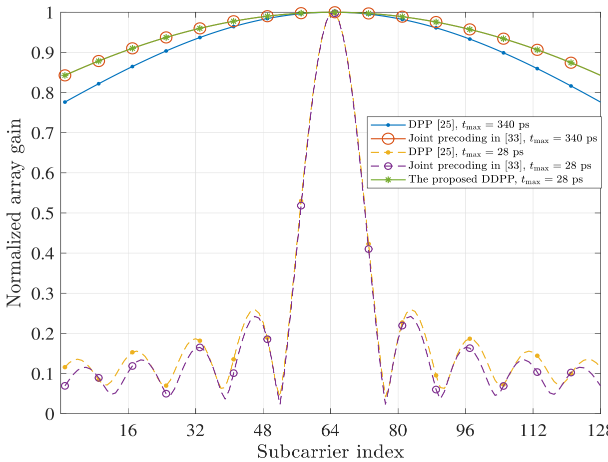


Fig. 6. Normalized array gain versus subcarrier index $m$ when $\sin(\theta_{l,u}) = 0.8$, $t_{\max} = 28$ ps, and $t_{\max} = 340$ ps.

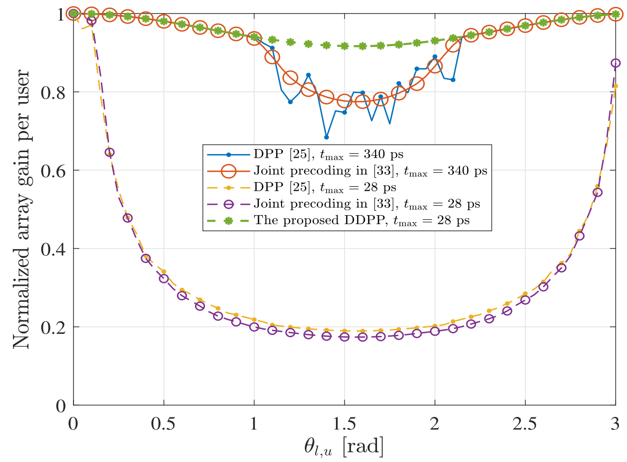


Fig. 7. Normalized array gain per user versus $\theta_{l,u}$ when $t_{\max} = 28$ and 340 ps.

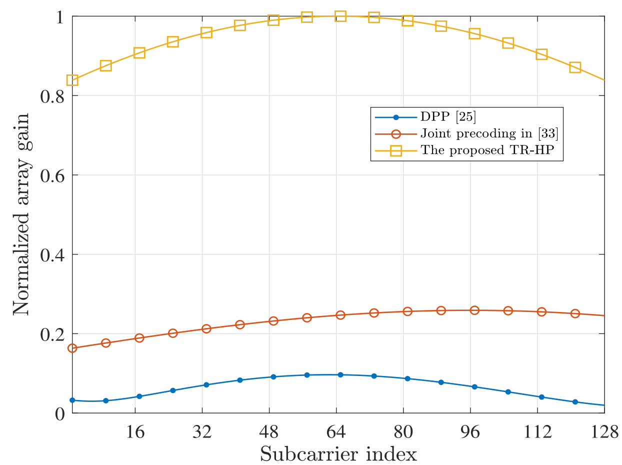


Fig. 8. Normalized array gain versus subcarrier index when $t_{\max} = 340$ ps, $t_{\text{res}} = 4$ ps, and $\sin(\theta_{l,u}) = 0.8$.

[25] drops to lower than 0.8 at margin subcarriers when $t_{\max} = 340$ ps. On the other hand, the normalized array gain of the hybrid precodings in [25] and [33] drops down fluctuating around 0.1 at non-central subcarriers when $t_{\max} = 28$ ps. However, the proposed DDPP maintains its robustness achieving a normalized array gain over 0.8 for all subcarriers.

To further evaluate the performance of DDPP compared to the reference works, Fig. 7 illustrates the normalized array gain per user versus $\theta_{l,u} \in [0, \pi]$ for $t_{\max} = 340$ and 28 ps. It is observed that the normalized array gain per user of the hybrid precodings in [25] and [33] declines as $\theta_{l,u}$ approaches $\pi/2$, where the normalized array gain per user is lower than 0.8 at $\theta_{l,u} = \pi/2$ when $t_{\max} = 340$ ps. Moreover, the normalized array gain per user of the hybrid precodings in [25] and [33] deteriorates to around 0.2 at $\theta_{l,u} = \pi/2$ when $t_{\max} = 28$ ps. This happens because the required time delay scales up as $\sin(\theta_{l,u})$ approaches 1 as shown in (17). Meanwhile, the proposed DDPP maintains a high degree of robustness against beam squint effect, where the normalized array gain per user is higher than 0.9 is guaranteed at all angles when $t_{\max} = 28$ ps. This superior performance is attributed to the proposed DDPP architecture, which reduces the maximum required delay range by a factor of $K - 1$.

### *B. Hybrid Analog Precoding Under Limited-Range and Finite-Resolution TTD Constraints*

In this subsection, we evaluate the peformance of the proposed TR-HP compared to the works in [25] and [33] when considering TTDs with limited-range and finite-resolution. Our main focus is to demonstrate the impact of time delay resolution, thus we consider practical TTDs with $t_{\max} = 340$ ps and $t_{\text{res}} = 4$ ps.

Fig. 8 illustrates the impact of time delay resolution on the normalized array gain over subcarriers for $t_{\max} = 340$ ps, $t_{\text{res}} = 4$ ps, and $\sin(\theta_{l,u}) = 0.8$. The results show that the precoding schemes in [25] and [33] suffer from a severe performance degradation due to the time delay resolution of TTDs, where the normalized array gain drops below 0.1 and 0.3 for all subcarriers, respectively. On the other hand, the proposed TR-HP achieves a normalized array gain higher than 0.8 at all subcarriers.

Fig. 9 shows the normalized array gain per user versus $\theta_{l,u}$ for TTDs with $t_{\max} = 340$ ps and $t_{\text{res}} = 4$ ps. It can be noted that the normalized array gain per user of the hybrid precoding in [25] drops below 0.1 at several angles, while the normalized array gain per user of the hybrid precoding in [33] fluctuates and declines to less than 0.2 at various angles. Nevertheless, the proposed TR-HP achieve normalized array gain very similar to that of the unconstrained DDPP precoder, with a normalized array gain per user greater than 0.9 at all angles.

Fig. 10 depicts the normalized array gain per user versus $\theta_{l,u}$ for the proposed HPP scheme at different fractional bandwidths $\beta = 0.1,\ 0.2,\ 0.3$ when $N_{\text{T}} = 256$, $t_{\max} = 340$ ps, and $t_{\text{res}} = 4$ ps. To adapt to increasing fractional bandwidths $\beta = \{0.1,\ 0.2,\ 0.3\}$, we increase the number of TTDs $K = \{16,\ 32,\ 64\}$, consistent with the linear relationship between the required number of TTDs and the fractional bandwidth [25]. It can be observed that the normalized array gain of the proposed HPP scheme maintains a robust performance $g(\theta_{l,u}) > 0.9$ at all angles for even higher

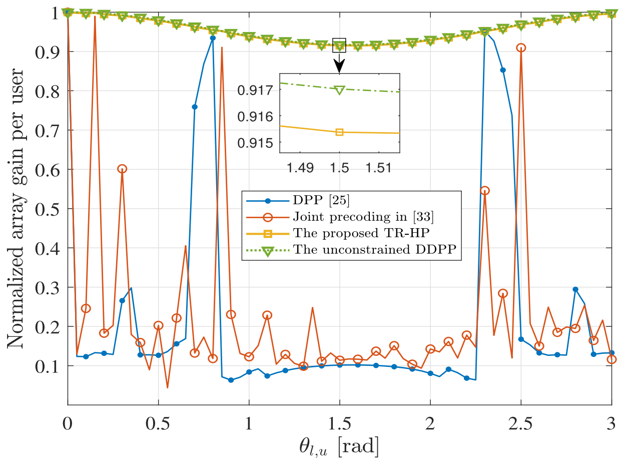


Fig. 9. Normalized array gain per user versus $\theta_{l,u}$ when $t_{\max} = 340$ ps and $t_{\rm res} = 4$ ps.

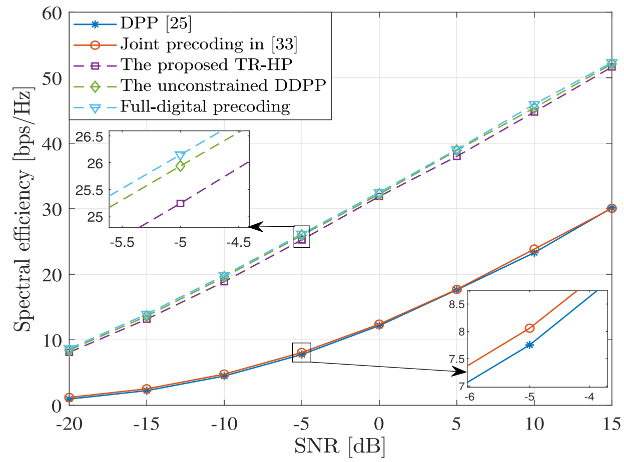


Fig. 11. Spectral efficiency versus $SNR$ when $t_{\max} = 340$ ps and $t_{\rm res} = 4$ ps.

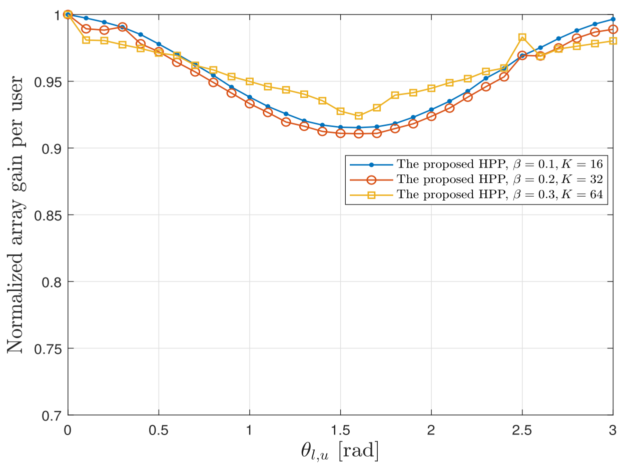


Fig. 10. Normalized array gain per user versus $\theta_{l,u}$ when $N_{\rm T} = 256$, $t_{\max} = 340$ ps, and $t_{\rm res} = 4$ ps.

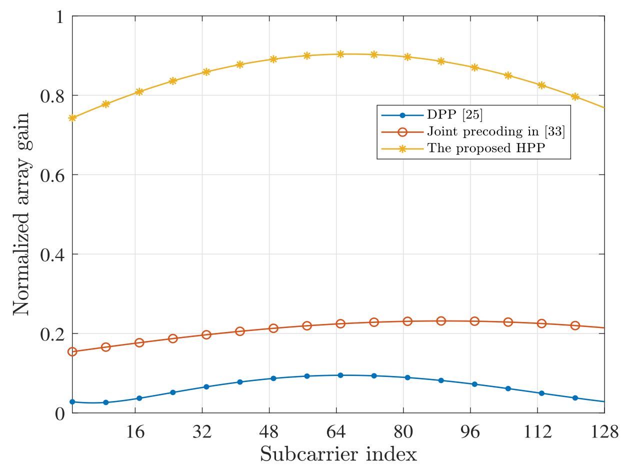


Fig. 12. Normalized array gain per user versus subcarrier index when $t_{\max} = 340$ ps, $t_{\rm res} = 4$ ps, $b = 2$ bits, and $\sin(\theta_{l,u}) = 0.8$.

fractional bandwidths up to 0.3 under the adoption of practical TTDs.

Fig. 11 shows the spectral efficiency performance when $U = N_{\rm s} = 4$, $t_{\max} = 340$ ps, and $t_{\rm res} = 4$ ps. It can be observed that the proposed TR-HP achieves a spectral efficiency of almost 97% of that achieved by the full-digital precoder, despite the limited-range and finite-resolution constraints of TTDs. Specifically, the spectral efficiency of the proposed TR-HP is almost 21 bps/Hz higher than that achieved by the hybrid precodings in [25] and [33] at $SNR = 10$ dB.

We conclude that the proposed TR-HP achieves angle-independent performance close to optimal under the hardware constraints of TTDs.

### C. Hybrid Precoding With Practical TTD and PS Constraints

In this subsection, we evaluate the performance of the proposed HPP and FD-HPP schemes under the consideration of the resolution constraint of PSs in addition to the TTDs constraints considered in the previous subsection. Particularly, we adopt low-resolution PSs that are digitally controlled by $b = 2$ bits, as well as limited-range and finite-resolution TTDs with $t_{\max} = 340$ ps and $t_{\rm res} = 4$ ps.

Fig. 12 illustrates the normalized array gain over subcarriers, considering limited-range and finite-resolution TTDs with $t_{\max} = 340$ ps, $t_{\rm res} = 4$ ps and low-resolution PSs controlled by $b = 2$ bits, with $\sin(\theta_{l,u}) = 0.8$. The results show that the hybrid precoding in [33] achieves a normalized array gain of around 0.2, while the hybrid precoding in [25] falls below 0.1 at all subcarriers. In contrast, the proposed HPP maintains a normalized array gain higher than 0.7 at all subcarriers.

Similar to Fig. 9, Fig. 13 depicts the normalized array gain per user versus $\theta_{l,u}$ for $t_{\max} = 340$ ps, $t_{\rm res} = 4$ ps, and $b = 2$ bits. The hybrid precodings in [25] and [33] exhibit a low normalized array gain per user of 0.1 at several angles. Meanwhile, the proposed HPP achieves a normalized array gain higher than 0.8 at all subcarriers, despite the hardware limitations of TTDs and PSs.

As the iterations represent the penalty of iterative methods, it is meaningful to study the effect of iteration number on the achieved performance. Fig. 14 depicts $\bar{g}/\bar{g}_{Iter}$ versus iteration number for different number of antennas, where $\bar{g}_{Iter}$ is the normalized array gain per multiple users at the last iteration. We assume $Iter = 10$, $t_{\max} = 28$ ps, $t_{\rm res} = 4$ ps, and $b = 2$ bits. It is observed that the proposed HPP has a fast

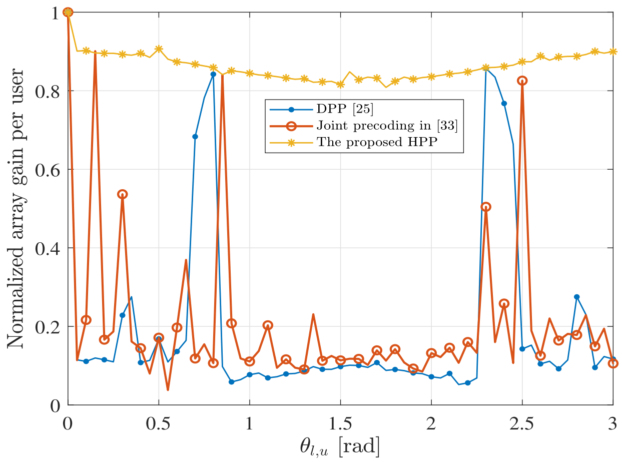


Fig. 13. Normalized array gain per user versus $\theta_{l,u}$ when $t_{\max} = 340$ ps, $t_{\text{res}} = 4$ ps, and $b = 2$ bits.

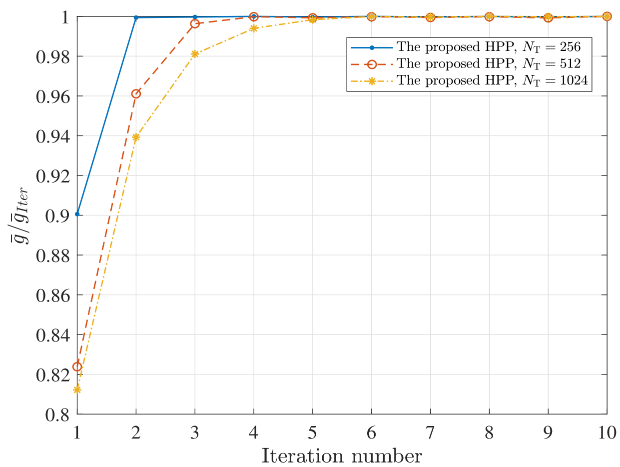


Fig. 14. Normalized array gain per multiple users versus iteration number when $t_{\max} = 28$ ps, $t_{\text{res}} = 4$ ps, and $b = 2$ bits.

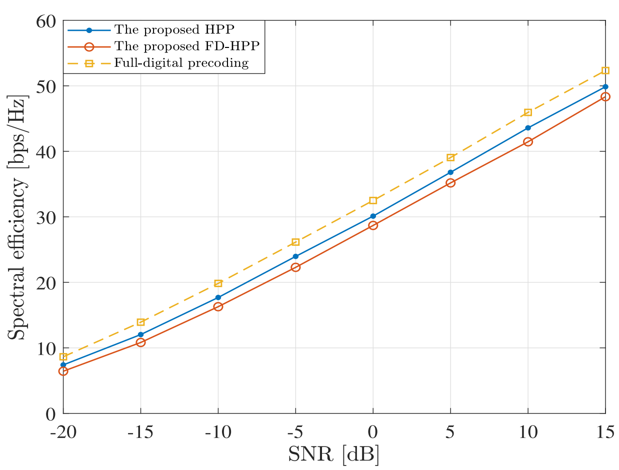


Fig. 15. Spectral efficiency versus $SNR$ when $t_{\max} = 28$ ps, $t_{\text{res}} = 4$ ps, and $b = 2$ bits.

convergence, where $\bar{g}/\bar{g}_{Iter} \geq 0.99$ is reached in 2, 4, and 5 iterations for $N_T = 256,\ 512,$ and 1024, respectively. This confirms the fast convergence and scalability of the proposed HPP, despite the hardware constraints of TTDs and PSs.

Fig. 15 depicts the spectral efficiency of the proposed HPP and FD-HPP precoding schemes assuming $U = N_s = 4$, $t_{\max} = 28$ ps, $t_{\text{res}} = 4$ ps, and $b = 2$ bits. It is observed that the spectral efficiency is only 2 bps/Hz lower than that achieved by the full-digital precoder at $SNR = 10$ dB. We conclude that the proposed HPP achieves near-optimal performance under the adoption of practical TTDs and PSs.

Similarly, the proposed FD-HPP achieves lower computational complexity at the cost of a slight performance reduction. Particularly, the spectral efficiency decreases by only 2 bps/Hz compared to HPP at $SNR = 10$ dB. In concise, the proposed FD-HPP proves to achieve a satisfactory performance with low computational overhead while adhering to the harware limitations of TTDs and PSs.

## VIII. Conclusion

In this work, we propose a novel hybrid precoding architecture along with several hardware-aware delay-phase precoding schemes that address the beam squint effect while complying with the hardware limitations of TTDs and PSs. Initially, we explore scenarios where TTDs are required for beam squint mitigation, focusing on the fractional bandwidth and the number of antennas. In this regard, we propose the DDPP architecture, which substantially alleviates the beam squint effect using practical, limited-range TTDs. Notably, the maximum delay range required for TTDs in DDPP is reduced by a factor of $(K-1)$ compared to the traditional DPP architecture. The DDPP architecture also demonstrates a high degree of scalability for UM-MIMO systems. Subsequently, we propose the TR-HP scheme that additionally addresses the time resolution constraint of TTDs. Specifically, we convert the time error resulting from the finite-resolution constraint of TTDs to the phase domain, where we compensate by introducing an equalization phase shift.

Afterwards, we include the resolution constraint of PSs, where we propose the HPP scheme that jointly optimizes TTDs and PSs under the consideration of their hardware limitations. In specific, we formulate the design of the hybrid analog precoder under TTDs and PSs constraints as a mixed-integer optimization problem and propose an alternating minimization-based iterative algorithm to solve this problem.

Eventually, with the aim of reducing the computational complexity of the HPP scheme, we propose the FD-HPP scheme, which ensures a fixed time delay value for all TTDs while maintaining lower computational overhead compared to the HPP scheme.

We further discover that the DDPP architecture consumes less power than the conventional DPP architecture, primarily due to the incorporation of practical, limited-range TTDs. In addition, our assessment of the computational complexity of the FD-HPP scheme relative to the HPP scheme reveals only a linear increase in complexity as a function of the fractional bandwidth and the number of antennas.

Simulation results validate the effectiveness of the proposed hybrid precoding schemes under the adoption of the DDPP architecture that utilizes practical TTDs and PSs. Specifically, near-optimal performance approaching that of the full-digital precoder is attained despite the limitations imposed by the hardware realizations of TTDs and PSs.

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## REFERENCES

[1] A. Najjar, M. El-Absi, and T. Kaiser, "Joint iterative delay-phase precoding for THz UM-MIMO with finite-resolution and limited-range true time delay," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2023, pp. 4311–4316.

[2] *The Next Hyper Connected Experience for All*, Samsung 6G Vision, Suwon-si, South Korea, Jun. 2020.

[3] W. Saad, M. Bennis, and M. Chen, "A vision of 6G wireless systems: Applications, trends, technologies, and open research problems," *IEEE Netw.*, vol. 34, no. 3, pp. 134–142, May 2020.

[4] F. Hu, Y. Deng, W. Saad, M. Bennis, and A. H. Aghvami, "Cellular-connected wireless virtual reality: Requirements, challenges, and solutions," *IEEE Commun. Mag.*, vol. 58, no. 5, pp. 105–111, May 2020.

[5] Z. Chen et al., "Terahertz wireless communications for 2030 and beyond: A cutting-edge frontier," *IEEE Commun. Mag.*, vol. 59, no. 11, pp. 66–72, Nov. 2021.

[6] J. C. Balzer et al., "THz systems exploiting photonics and communications technologies," *IEEE J. Microw.*, vol. 3, no. 1, pp. 268–288, Jan. 2023.

[7] M. El-Absi, A. A. Abbas, A. Abuelhaija, F. Zheng, K. Solbach, and T. Kaiser, "High-accuracy indoor localization based on chipless RFID systems at THz band," *IEEE Access*, vol. 6, pp. 54355–54368, 2018.

[8] J. Kokkoniemi, J. Lehtomäki, and M. Juntti, "A discussion on molecular absorption noise in the terahertz band," *Nano Commun. Netw.*, vol. 8, pp. 35–45, Jun. 2016.

[9] I. F. Akyildiz, C. Han, and S. Nie, "Combating the distance problem in the millimeter wave and terahertz frequency bands," *IEEE Commun. Mag.*, vol. 56, no. 6, pp. 102–108, Jun. 2018.

[10] H. Sarieddeen, M.-S. Alouini, and T. Y. Al-Naffouri, "Terahertz-band ultra-massive spatial modulation MIMO," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 9, pp. 2040–2052, Sep. 2019.

[11] M. R. Akdeniz et al., "Millimeter wave channel modeling and cellular capacity evaluation," *IEEE J. Sel. Areas Commun.*, vol. 32, no. 6, pp. 1164–1179, Jun. 2014.

[12] C. Han, A. O. Bicen, and I. F. Akyildiz, "Multi-wideband waveform design for distance-adaptive wireless communications in the terahertz band," *IEEE Trans. Signal Process.*, vol. 64, no. 4, pp. 910–922, Feb. 2016.

[13] I. F. Akyildiz and J. M. Jornet, "Realizing ultra-massive MIMO ($1024 \times 1024$) communication in the (0.06–10) terahertz band," *Nano Commun. Netw.*, vol. 8, pp. 46–54, Jun. 2016.

[14] W. Roh et al., "Millimeter-wave beamforming as an enabling technology for 5G cellular communications: Theoretical feasibility and prototype results," *IEEE Commun. Mag.*, vol. 52, no. 2, pp. 106–113, Feb. 2014.

[15] D. Ha, K. Lee, and J. Kang, "Energy efficiency analysis with circuit power consumption in massive MIMO systems," in *Proc. IEEE 24th Annu. Int. Symp. Pers., Indoor, Mobile Radio Commun. (PIMRC)*, Sep. 2013, pp. 938–942.

[16] S. Han, I. Chih-Lin, Z. Xu, and C. Rowell, "Large-scale antenna systems with hybrid analog and digital beamforming for millimeter wave 5G," *IEEE Commun. Mag.*, vol. 53, no. 1, pp. 186–194, Jan. 2015.

[17] A. Najjar, M. El-Absi, and T. Kaiser, "Dynamic array of subarrays architecture based hybrid beamforming in UM-MIMO systems," in *Proc. 4th Int. Workshop Mobile THz Syst. (IWMTS)*, Jul. 2021, pp. 1–5.

[18] C. Han, L. Yan, and J. Yuan, "Hybrid beamforming for terahertz wireless communications: Challenges, architectures, and open problems," *IEEE Wireless Commun.*, vol. 28, no. 4, pp. 198–204, Aug. 2021.

[19] Y. Chen and C. Han, "Razor-sharp narrowbeam communications and management for terahertz wireless networks (invited paper)," in *Proc. Comput., Commun. IoT Appl. (ComComAp)*, Nov. 2021, pp. 335–340.

[20] H. Hashemi, T.-S. Chu, and J. Roderick, "Integrated true-time-delay-based ultra-wideband array processing," *IEEE Commun. Mag.*, vol. 46, no. 9, pp. 162–172, Sep. 2008.

[21] C. Lin, G. Y. Li, and L. Wang, "Subarray-based coordinated beamforming training for mmWave and sub-THz communications," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 9, pp. 2115–2126, Sep. 2017.

[22] S. A. Busari et al., "Generalized hybrid beamforming for vehicular connectivity using THz massive MIMO," *IEEE Trans. Veh. Technol.*, vol. 68, no. 9, pp. 8372–8383, Sep. 2019.

[23] J. Tan and L. Dai, "THz precoding for 6G: Challenges, solutions, and opportunities," *IEEE Wireless Commun.*, vol. 30, no. 4, pp. 132–138, Aug. 2023, doi: 10.1109/MWC.015.2100674.

[24] J. Tan and L. Dai, "Delay-phase precoding for THz massive MIMO with beam split," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2019, pp. 1–6.

[25] L. Dai, J. Tan, Z. Chen, and H. V. Poor, "Delay-phase precoding for wideband THz massive MIMO," *IEEE Trans. Wireless Commun.*, vol. 21, no. 9, pp. 7271–7286, Sep. 2022.

[26] Y. Wu, G. Song, H. Liu, L. Xiao, and T. Jiang, "3-D hybrid beamforming for terahertz broadband communication system with beam squint," *IEEE Trans. Broadcast.*, vol. 69, no. 1, pp. 264–275, Mar. 2023.

[27] T. Maiwald et al., "A review of integrated systems and components for 6G wireless communication in the D-band," *Proc. IEEE*, vol. 111, no. 3, pp. 220–256, Mar. 2023.

[28] A. Karakuzulu, M. H. Eissa, D. Kissinger, and A. Malignaggi, "Broadband 110–170 GHz true time delay circuit in a 130-nm SiGe BiCMOS technology," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Aug. 2020, pp. 775–778.

[29] A. Karakuzulu, W. A. Ahmad, D. Kissinger, and A. Malignaggi, "A four-channel bidirectional D-band phased-array transceiver for 200 Gb/s 6G wireless communications in a 130-nm BiCMOS technology," *IEEE J. Solid-State Circuits*, vol. 58, no. 5, pp. 1310–1322, May 2023.

[30] L. Steinweg, C. Carta, and F. Ellinger, "A hybrid true-time and phase-delayed approach for millimeter-wave beam steering," *IEEE Trans. Microw. Theory Techn.*, vol. 70, no. 9, pp. 4318–4327, Sep. 2022.

[31] C. Maurette-Blasini, K. Kuliabin, S. Chartier, and R. Quay, "An H-band mHEMT-based millimeter-wave true-time delay MMIC," in *Proc. 14th German Microw. Conf. (GeMiC)*, May 2022, pp. 96–99.

[32] M. W. Mansha, A. Elmenshawi, S. Muralidharan, and M. M. Hella, "A W-band digitally controlled bidirectional transmission line phase shifter in 22-nm FDSOI," *IEEE Microw. Wireless Technol. Lett.*, vol. 33, no. 7, pp. 1003–1006, Mar. 2023.

[33] D. Q. Nguyen and T. Kim, "Joint hybrid delay-phase precoding under true-time delay constraints in wideband THz massive MIMO systems," 2023, *arXiv:2212.07484v4*.

[34] L. Yan, C. Han, and J. Yuan, "Energy-efficient dynamic-subarray with fixed true-time-delay design for terahertz wideband hybrid beamforming," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 10, pp. 2840–2854, Oct. 2022.

[35] A. Faisal, H. Sarieddeen, H. Dahrouj, T. Y. Al-Naffouri, and M.-S. Alouini, "Ultramassive MIMO systems at terahertz bands: Prospects and challenges," *IEEE Veh. Technol. Mag.*, vol. 15, no. 4, pp. 33–42, Dec. 2020.

[36] E. Ghaderi, A. S. Ramani, A. A. Rahimi, D. Heo, S. Shekhar, and S. Gupta, "An integrated discrete-time delay-compensating technique for large-array beamformers," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 66, no. 9, pp. 3296–3306, Sep. 2019.

[37] B. Zhai, Y. Zhu, A. Tang, and X. Wang, "THzPrism: Frequency-based beam spreading for terahertz communication systems," *IEEE Wireless Commun. Lett.*, vol. 9, no. 6, pp. 897–900, Jun. 2020.

[38] A. Najjar, M. El-Absi, and T. Kaiser, "Non-overlapped subarrays based wideband delay-phase hybrid beamforming," in *Proc. 5th Int. Workshop Mobile THz Syst. (IWMTS)*, Jul. 2022, pp. 1–5.

[39] L. Yan, C. Han, T. Yang, and J. Yuan, "Dynamic-subarray with fixed-true-time-delay architecture for terahertz wideband hybrid beamforming," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2021, pp. 1–6.

[40] D. Q. Nguyen and T. Kim, "Joint delay and phase precoding under true-time delay constraints for THz massive MIMO," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2022, pp. 3496–3501.

[41] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2004.

[42] J. W. Brown and R. V. Churchill, *Complex Variables and Applications*, 8th ed. Boston, MA, USA: McGraw-Hill, 2009.

[43] O. E. Ayach, S. Rajagopal, S. Abu-Surra, Z. Pi, and R. W. Heath, "Spatially sparse precoding in millimeter wave MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 13, no. 3, pp. 1499–1513, Mar. 2014.

[44] T. S. Rappaport, G. R. MacCartney, M. K. Samimi, and S. Sun, "Wideband millimeter-wave propagation measurements and channel models for future wireless communication system design," *IEEE Trans. Commun.*, vol. 63, no. 9, pp. 3029–3056, Sep. 2015.

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